

Practical Applications of Whole Genome Sequencing and Bioinformatics in Galway Reference Laboratory Services

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Galway Reference Laboratory Service.

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HSE West and North West



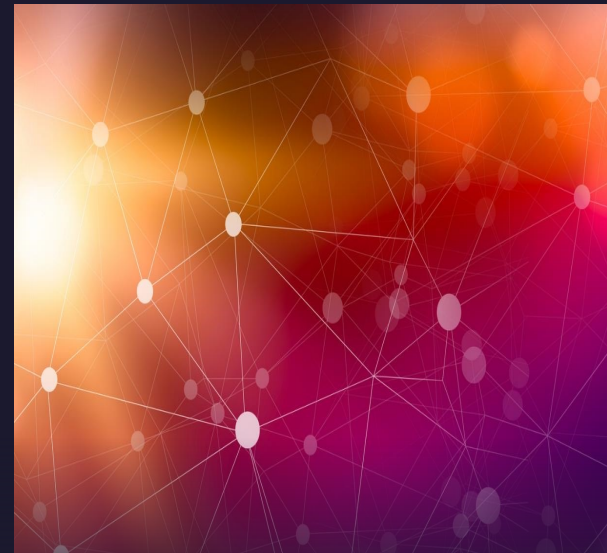
Galway
University
Hospitals
Óspidéal na n-Oileánach Galway
University Hospitals Galway
Hospitals Galway University Hospitals

Who Are We?

National *Salmonella*, *Shigella* and *Listeria*
Reference Laboratory, NSSLRL
and

National CPE Reference Laboratory, NCPERL

Based in the Microbiology dept. on the campus of
University Hospital Galway



What do we do ?

Whole Genome Sequencing (WGS) and analysis of :

- All *Salmonella*, *Shigella* and *Listeria* isolated from humans.
- All CPE from diagnostic isolates and large proportion of CPE isolates from screening samples, also from humans.
 - Limited numbers of Carbapenem Resistant *Acinetobacter baumannii* and Carbapenem Resistant *Pseudomonas aeruginosa*.

These activities adds up to approximately:

2000 CPE/CPO isolates

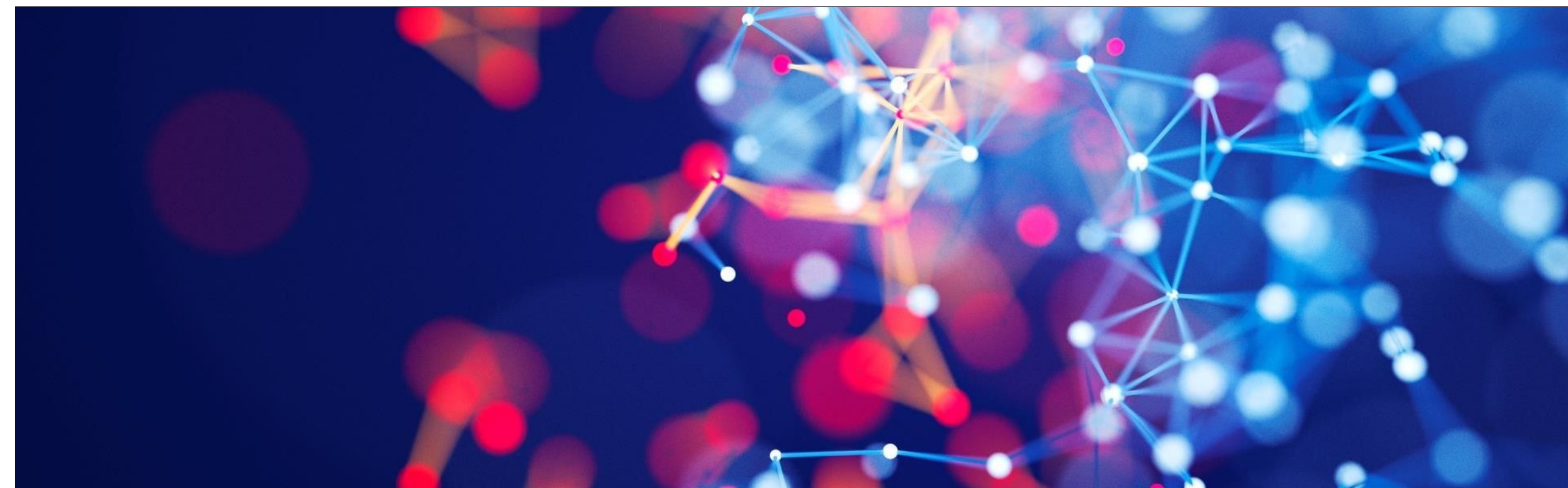
140 *Shigella* Isolates

400 *Salmonella* Isolates

10 to 20 *Listeria* Isolates

What do we do ?

- Support clinical laboratories by confirming and subtyping *Salmonella*, *Shigella* and *Listeria* and CPE isolates and reporting AMR determinants of the isolates.
- Provide national data to inform public health on the scope issue/outbreak and the effectiveness of responses.
- Liaise with HPSC and the European Centre for Disease Control (ECDC) to help identify and control international dimensions to outbreaks.
- Liaise with the Department of Agriculture and the Marine (DAFM) to help identify and control national outbreaks.
 - This involves sharing sequencing data without any patient identifying data. This information is primarily to guide public health intervention and to control transmission of infection



Technology



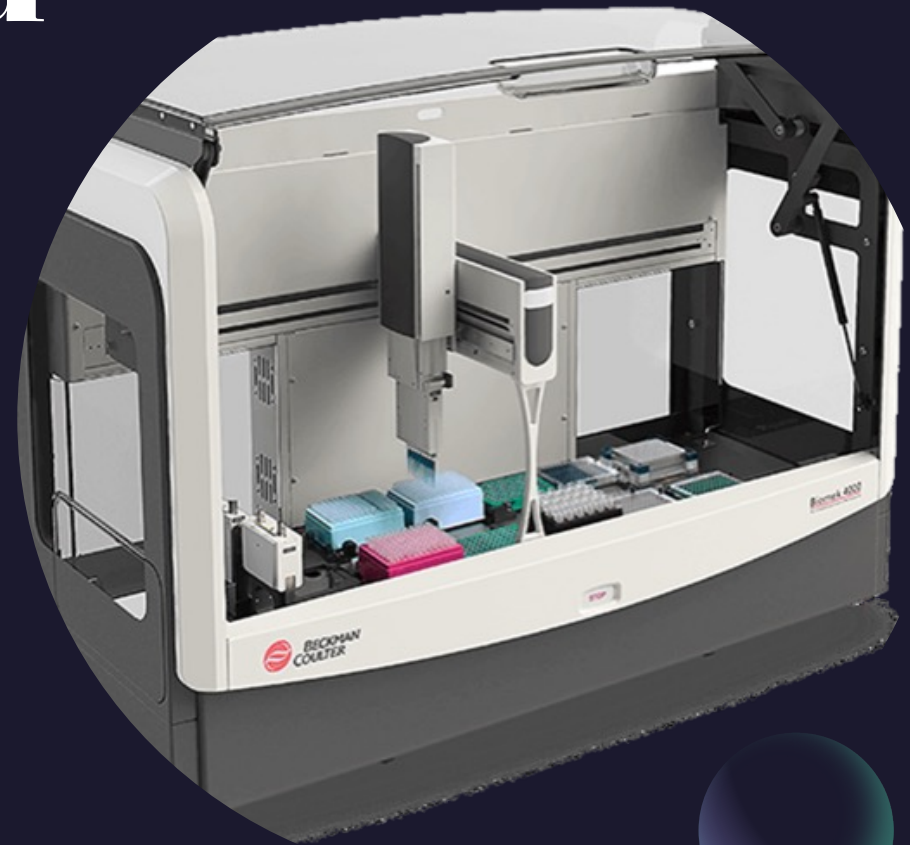
DNA Extraction and Purification

Qiagen EZ2
Connect MDx
EZ1/EZ2 Tissue Kit

This is a cartridge based system that utilises magnetic beads for nucleic acid extraction and purification. The cartridges are single use and there are separate robotics for each channel, this helps avoid contamination.

Robotic Liquid Handling

Instrument: Biomek4000
Library Preparation kit:
Illumina DNA Prep



This instrument performs the library preparation for our sequencing runs. Quality of the libraries are checked using Qbit DNA quantification and by using the Agilent Tapestation to determine the fragment size.



Sequencing

Instrument:

Illumina

NextSeq1000

Kit:

X-LEAP P1 600 Cycle

Once the libraries have passed the quality checks they are then loaded on the NextSeq. Sequencing takes approximately 34 hours.



Bruker MBioSEQ™
Ridom Typer

Sequencing Analysis

Quality trimming and De novo assembly (SKESA/SPAdes)

Quality scoring (FastQC, N50, coverage contamination check)

Geno-Serotyping/Serogrouping, MLST

cgMLST/pgMLST and accessory genome analysis (for determining relatedness of isolates)

AMR detection (NCBI AMRFinderPlus) for acquired resistance and point mutations

Virulence gene detection for certain pathogens

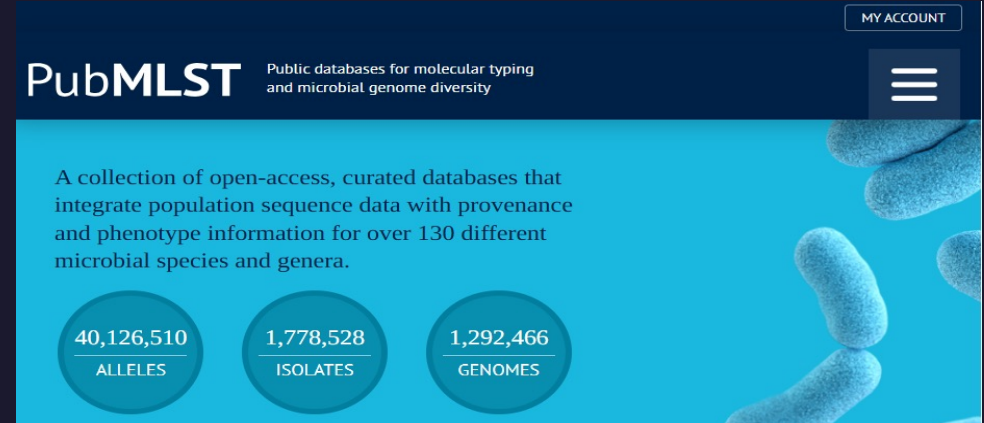
Bioinformatic software and tools used in Galway Reference Laboratory Service



The banner features the Bruker logo on the left and the Ridom logo on the right, which consists of four colored circles (A, C, G, T) arranged in a square. The background is a blue-toned image of a DNA double helix with binary code (0s and 1s) overlaid. The text "Bruker MBioSEQ™ Ridom Typer" is prominently displayed in the center.

BRUKER | **ridom**

Bruker MBioSEQ™ Ridom Typer

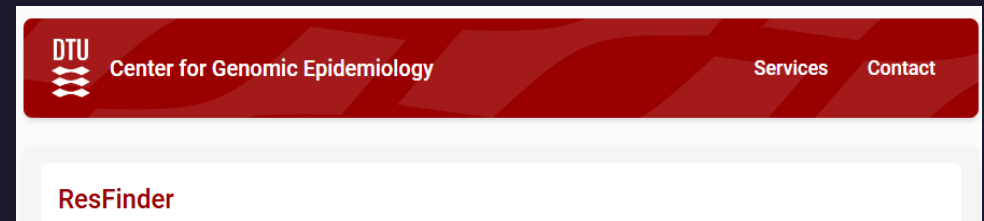


The banner has a dark blue header with the "PubMLST" logo and the tagline "Public databases for molecular typing and microbial genome diversity". A "MY ACCOUNT" link and a hamburger menu icon are in the top right. The main content area is light blue and features a paragraph about open-access databases. Below this, three circular statistics are shown: 40,126,510 ALLELES, 1,778,528 ISOLATES, and 1,292,466 GENOMES. The background on the right shows a microscopic view of bacteria.

PubMLST Public databases for molecular typing and microbial genome diversity

A collection of open-access, curated databases that integrate population sequence data with provenance and phenotype information for over 130 different microbial species and genera.

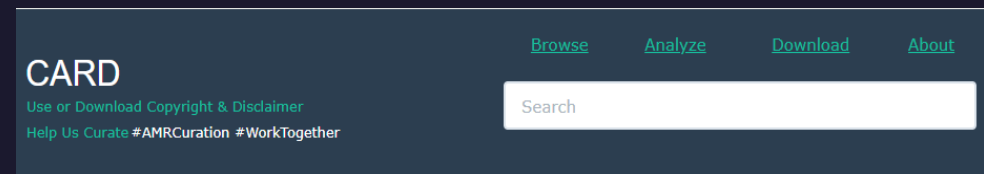
40,126,510 ALLELES	1,778,528 ISOLATES	1,292,466 GENOMES
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The banner has a red header with the DTU logo and the text "Center for Genomic Epidemiology". Links for "Services" and "Contact" are on the right. Below the header is a white box with the "ResFinder" logo.

DTU Center for Genomic Epidemiology [Services](#) [Contact](#)

ResFinder



The banner has a dark blue header with the "CARD" logo. Links for "Browse", "Analyze", "Download", and "About" are on the right. Below the header is a white box with the text "Use or Download Copyright & Disclaimer" and "Help Us Curate #AMRCuration #WorkTogether". A search bar is on the right.

CARD [Browse](#) [Analyze](#) [Download](#) [About](#)

Use or Download Copyright & Disclaimer
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Search



The banner features the "Enterobase" logo in a bold, sans-serif font. The background is a light blue image of a cell or tissue.

Enterobase



The banner features the "Institut Pasteur" logo in a serif font, with "BIGSdb-Pasteur" in a smaller font below it. The background is a light blue image of a cell or tissue.

**INSTITUT
pasteur**
BIGSdb-Pasteur



The banner features the "SeqSero2/SeqSero2S" logo in a bold, sans-serif font. Below it is the text "Salmonella Serotyping by Whole Genome Sequencing". The background is a light blue image of a cell or tissue.

SeqSero2/SeqSero2S
Salmonella Serotyping by Whole Genome Sequencing

Cluster Designation

All isolates are analysed with other isolates of the same CC/ST/MLST type using core genome MLST, +/- Accessory genome or by pan genome MLST, these differences are known as Allelic Differences or AD's

- Ones that are found to be closely related are given a cluster designation
- When an isolate is part of an identified cluster this is indicated on the report returned to the sender of that isolate.
- Clusters are designated by an alpha-numeric string in which the first one or two characters are letters related to the species, the first two numerals indicate the year and the final three numerals indicate the order of identification of clusters in that years. For example S24-003 is the third cluster of *Salmonella enterica* identified in 2024.

Listeria monocytogenes

2 or more isolates with 0 to 7 AD is designated a cluster.

Subsequent isolates are added to the cluster if they are 0 to 7 AD from any isolate previously designated as belonging to the cluster. No time limits are applied to initial definition of *L. monocytogenes* clusters as there is significant potential for low numbers of cases dispersed in time to be epidemiologically linked.

Shigella species

2 or more isolates with 0 to 5 AD within a 100 days is designated a cluster.

Subsequent isolates are added to the cluster if they are no more than 5 AD from any isolate previously designated as belonging to the cluster

Salmonella enterica

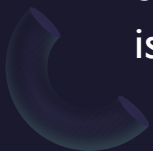
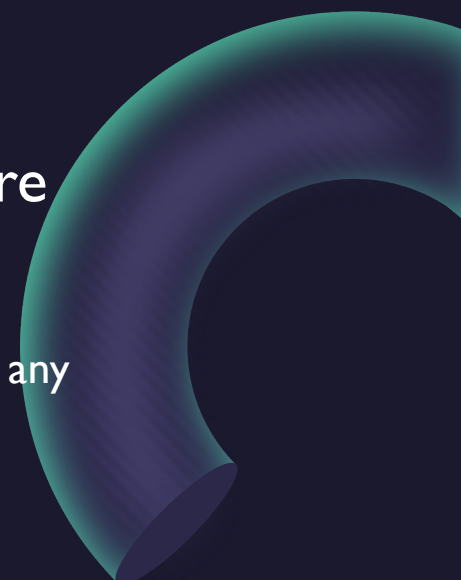
2 or more isolates with 0 to 7 AD within 100 days or 5 or more isolates with 0 to 7 AD within 2 years is designated a cluster.

- Subsequent isolates are added to the cluster if they are no more than 7 AD from any isolate previously designated as belonging to the cluster.

Salmonella enterica serovar Enteritidis

2 or more isolates with 0 to 3 AD within 100 days is designated a cluster.

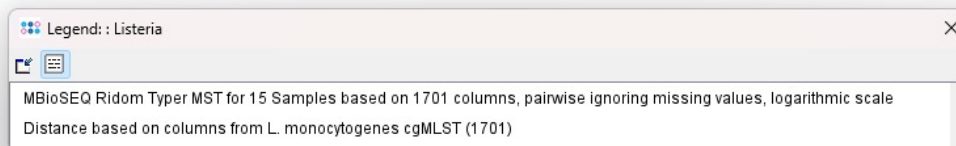
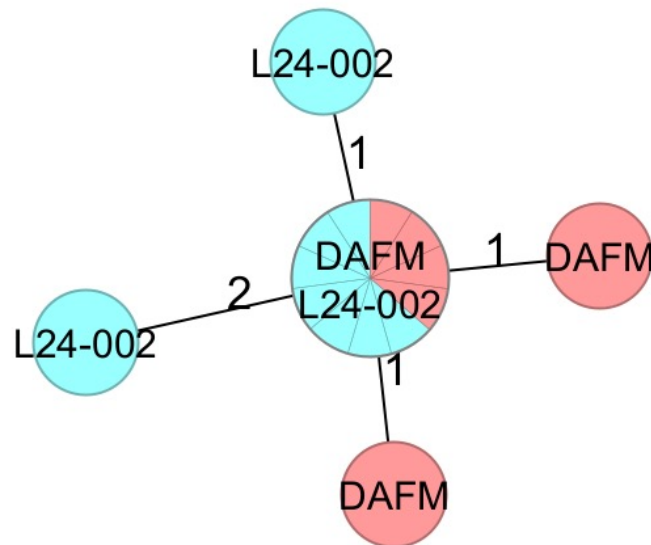
- Subsequent isolates are added to the cluster if they are no more than 3 AD from any isolate previously designated as belonging to the cluster



CPE, CPO, ESBL and other Enterobacterales

- 3 or more isolates from one institution or 5 or more isolates from 2 or more institutions with 0 to 10 AD within 100 days is designated a cluster.
- 10 or more isolates with 0 to 10 AD within 2 years is designated a cluster.
- Subsequent isolates are added to the cluster if they are no more than 10 AD from any isolate previously designated as belonging to the cluster
- For CPE and CPO 2 isolates from one institution with 0 to 10 AD is not designated a cluster but the report on the second isolate will include the comment “This isolate is indistinguishable from/closely related to isolate [Insert number] from your laboratory].”

Listeria Outbreak 2024-2025



L24-002 Total 9 isolates from 8 patients, 1 had LM isolated from CSF and blood

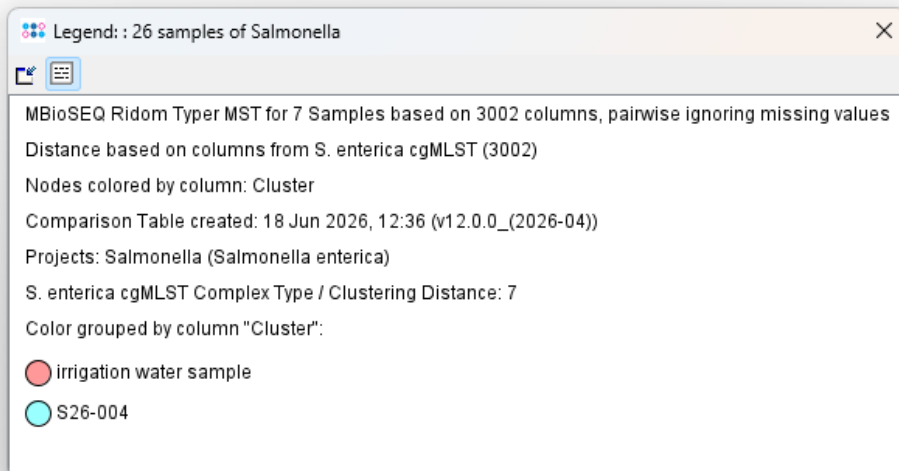
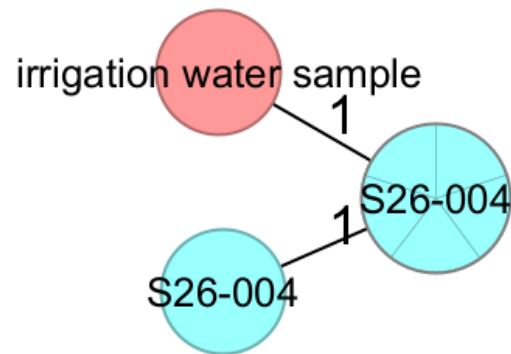
In 2024 we 4 isolates

Cluster designated after 2nd isolate found, HPSC informed

In 2025 we received 5 isolates

- Query with DAFM- have they seen this strain?
- When they found a match they sent us files also for clustering
- Linked back to single food business operator
- Listeria found in their “Ready to Heat” meals
- Recall on foods,
- FBO voluntarily shut down manufacturing for deep cleaning
- No further cases have been identified, last case was 10/07/2025

S. Bovismorbificans Outbreak - Ongoing



- 1st patient detected on 09/02, 2nd patient 30/03- cluster declared- HPSC informed
- 6 Patients affected to date
- FSAI received a query as to whether we had seen S.Bovismorbificans as it had been isolated Irrigation water from seed sprouting facility
- Irrigation water sequence from seed sprouting facility received 10/04 – clustered with human cases
- HPSC informed 16th of April 2026 Ireland posted an EpiPulse event
- As of 10/06/2026 11 countries have had 106 matching cases confirmed via sequencing

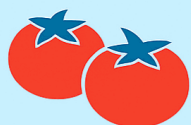
Long Term International Outbreak

 Eurosurveillance

Recurring multi-country outbreaks of *Salmonella* Strathcona linked to tomatoes, Europe, 2011–2024



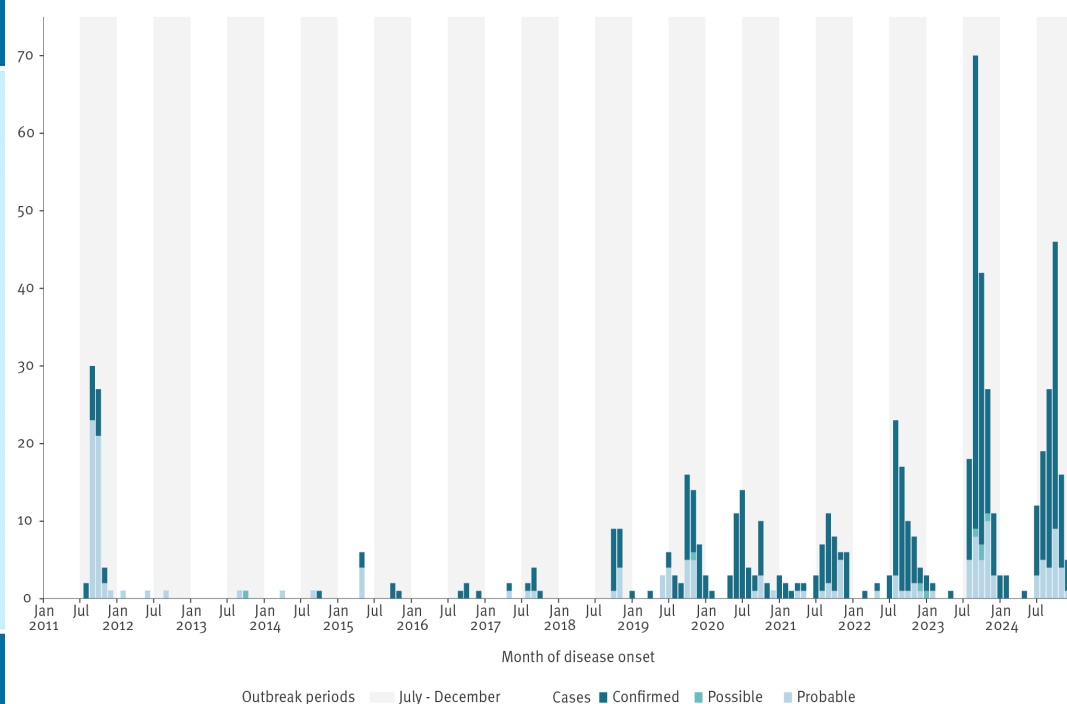
643 cases
across 17 countries



Small tomatoes
identified as the
likely vehicle



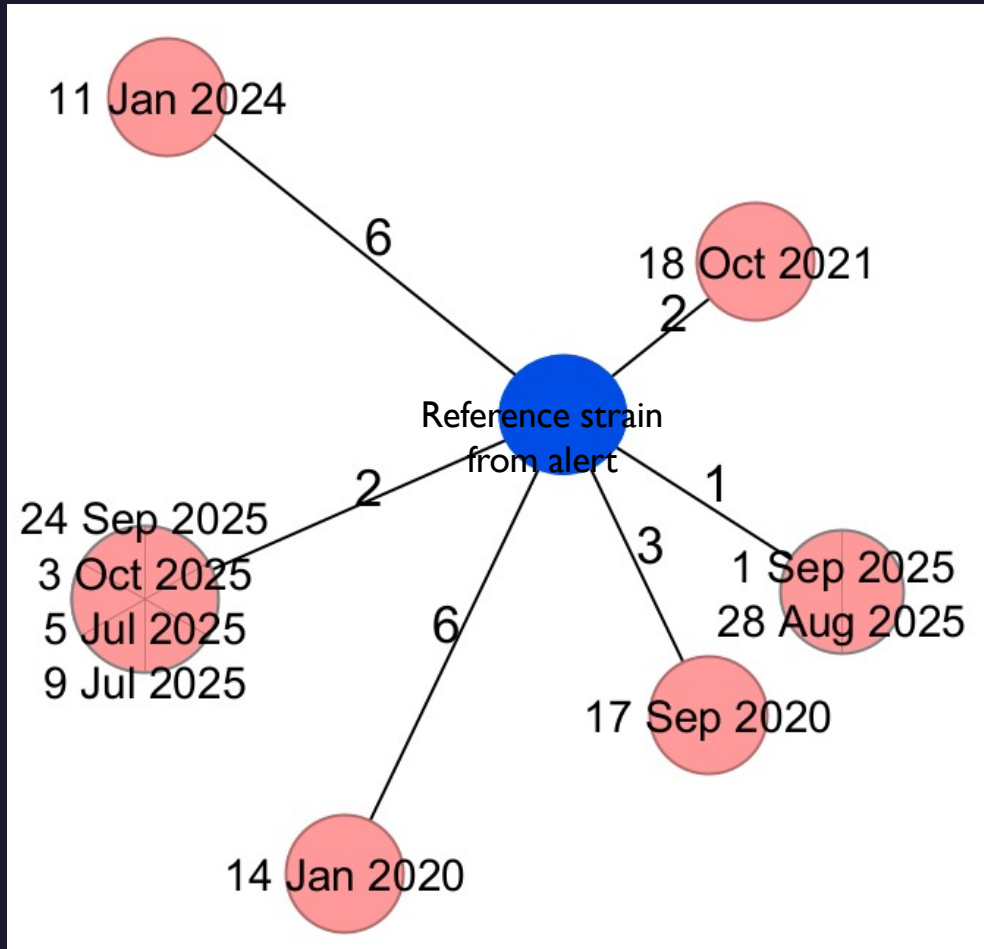
Seasonal pattern
of recurrence



Brait et al., Euro Surveill. 2025;30(41)

On 27 October 2023, Germany posted a notification on the European surveillance portal for infectious diseases EpiPulse, reporting on 46 cases of *S. Strathcona* occurring since August 2023. Several other EU/EEA countries also reported cases of *S. Strathcona* infections during the same period and subsequently, the outbreak was investigated by the affected countries. This cluster has continued into 2025

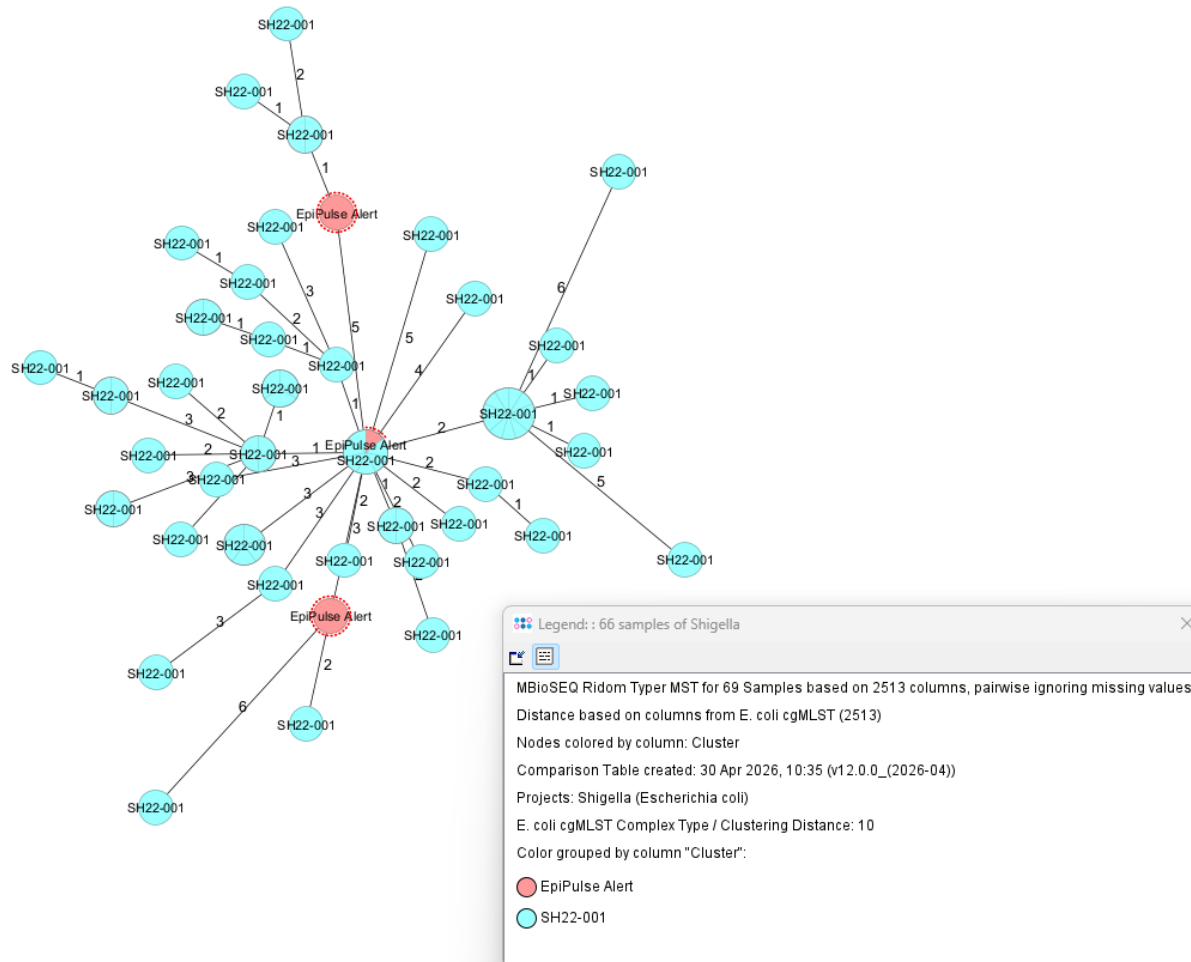
Irish isolates



Characteristics	Confirmed case (n = 469) ^a		Probable case (n = 161) ^b		Possible case (n = 13) ^c		Outbreak case (n = 643)		Non-outbreak case (n = 19) ^d		Sequenced isolates (n = 496)		Total number of cases (n = 662)	
Allelic difference (AD)														
Distance	0–7 AD		NA		8–13 AD		≤13 AD or a probable case		> 13 AD		NA			
Time frame	2023–2024	2011–2022	2023–2024	2011–2022	2023–2024	2011–2022	2023–2024	2011–2022	2023–2024	2011–2022	2023–2024	2011–2022	2023–2024	2011–2022
Total number of cases	243	226	58	103	5	8	306	337	2	17	250	246	308	354
Country														
Austria	47	24	6	0	0	0	53	24	0	0	71		77	
Belgium	3	3	0	0	0	0	3	3	0	2	8		8	
Croatia	0	0	6	0	0	0	6	0	0	0	0		6	
Czechia	12	4	4	0	2	0	18	4	0	0	18		22	
Denmark	9	42	0	39	1	2	10	83	0	0	54		93	
England	28	13	4	2	2	0	34	15	1	2	46		52	
Finland	4	0	0	0	0	2	4	2	0	0	4		6	
France	19	9	13	14	0	0	32	23	0	0	28		55	
Germany	85	88	11	44	0	1	96	133	1	13	188		243	
Ireland	1	3	0	0	0	0	1	3	0	0	4		4	
Luxembourg	3	4	0	0	0	0	3	4	0	0	7		7	
The Netherlands	2	1	0	0	0	0	2	1	0	0	3		3	
Norway	2	3	1	0	0	0	3	3	0	0	5		6	
Scotland	0	0	5	0	0	2	5	2	0	0	0		7	
Slovenia	8	1	2	0	0	0	10	1	0	0	9		11	
Sweden	2	0	4	0	0	1	6	1	0	0	2		7	
Switzerland	18	31	2	4	0	0	20	35	0	0	49		55	

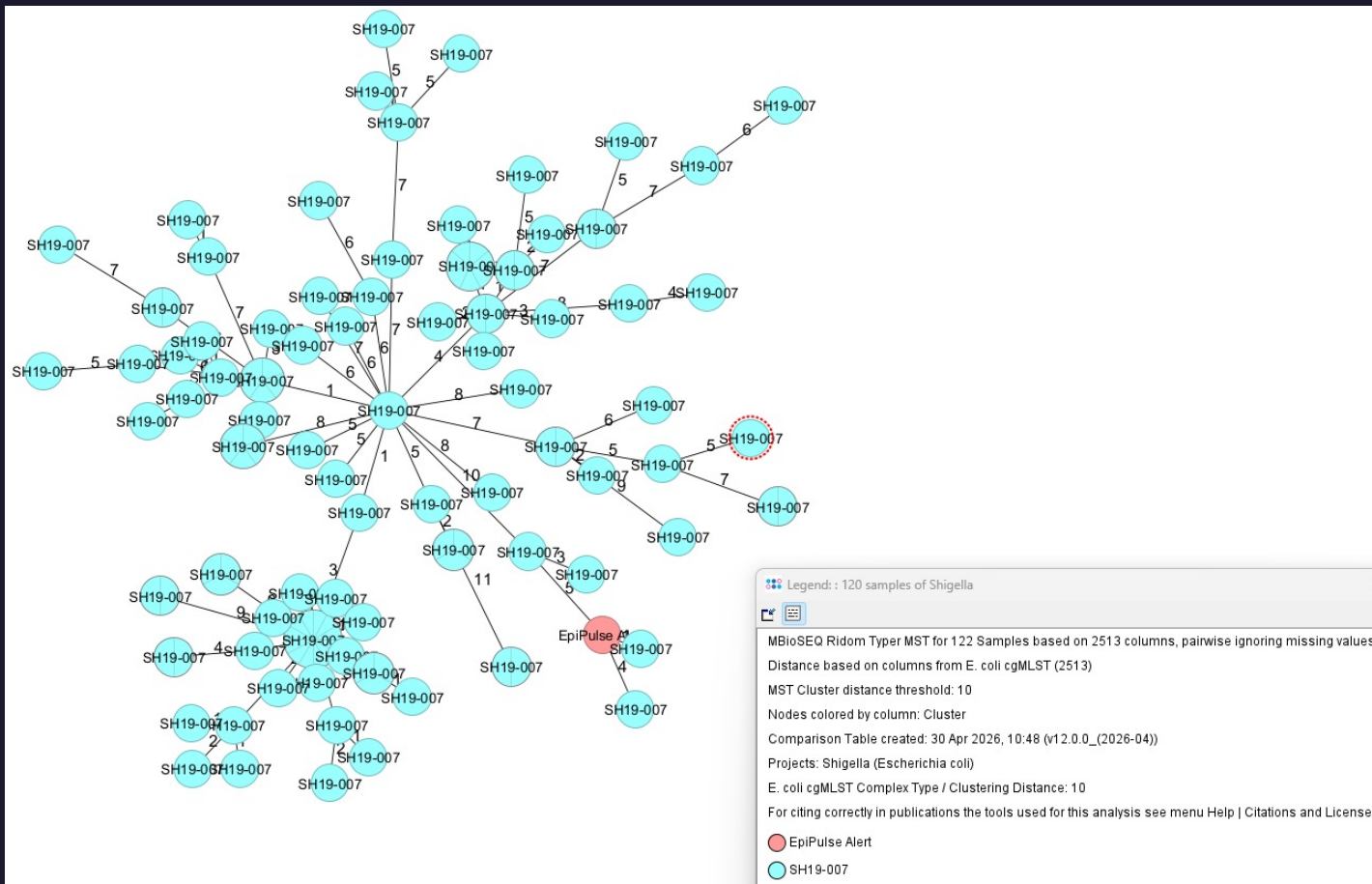
Only 4 Irish isolates were included in this paper, we have subsequently had 8 more isolates identified in 2025.

Shigella international clusters



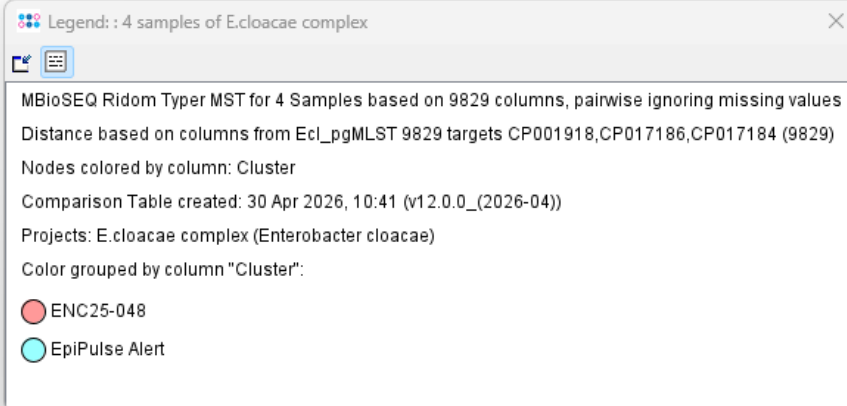
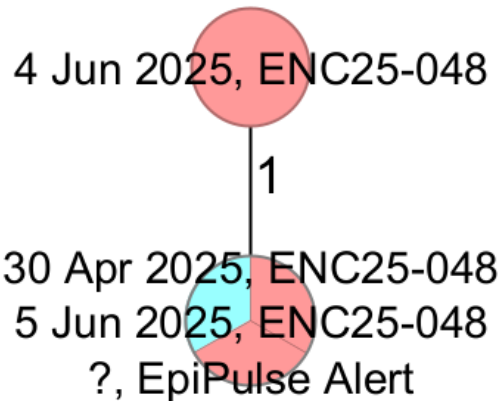
- SH22-001 1st isolate in 2022, still seeing isolates today. The isolates steadily gain differences
- Occasionally the ECDC have to merge clusters due to similarity- this practice is especially prevalent with Shigella as the clusters evolve over time.
- We are unique in that we sequence all Shigella so we have seen the evolution of the clusters in real time.

1st CPE Shigella in Europe



- ST245 *S.flexneri* harbouring *blaOXA-181* on the *incX3* plasmid (long read sequencing performed on this isolate)
- Part of a cluster that has been circulating in Ireland since 2017, to date the only isolate with *blaOXA-181*
- *Other resistance genes:*
 - *blaOXA-1*, *catA1*, *QnrS1*, *tet(B)* and *dfrA1*
 - *Common resistance mechanisms in this cluster* *blaOXA-1*, *catA1*, *tet(B)* and *dfrA1*
- An Enterobase search has revealed this cluster has been seen 10 countries within the HC10 hierarchical clustering threshold.

International CPE Clusters



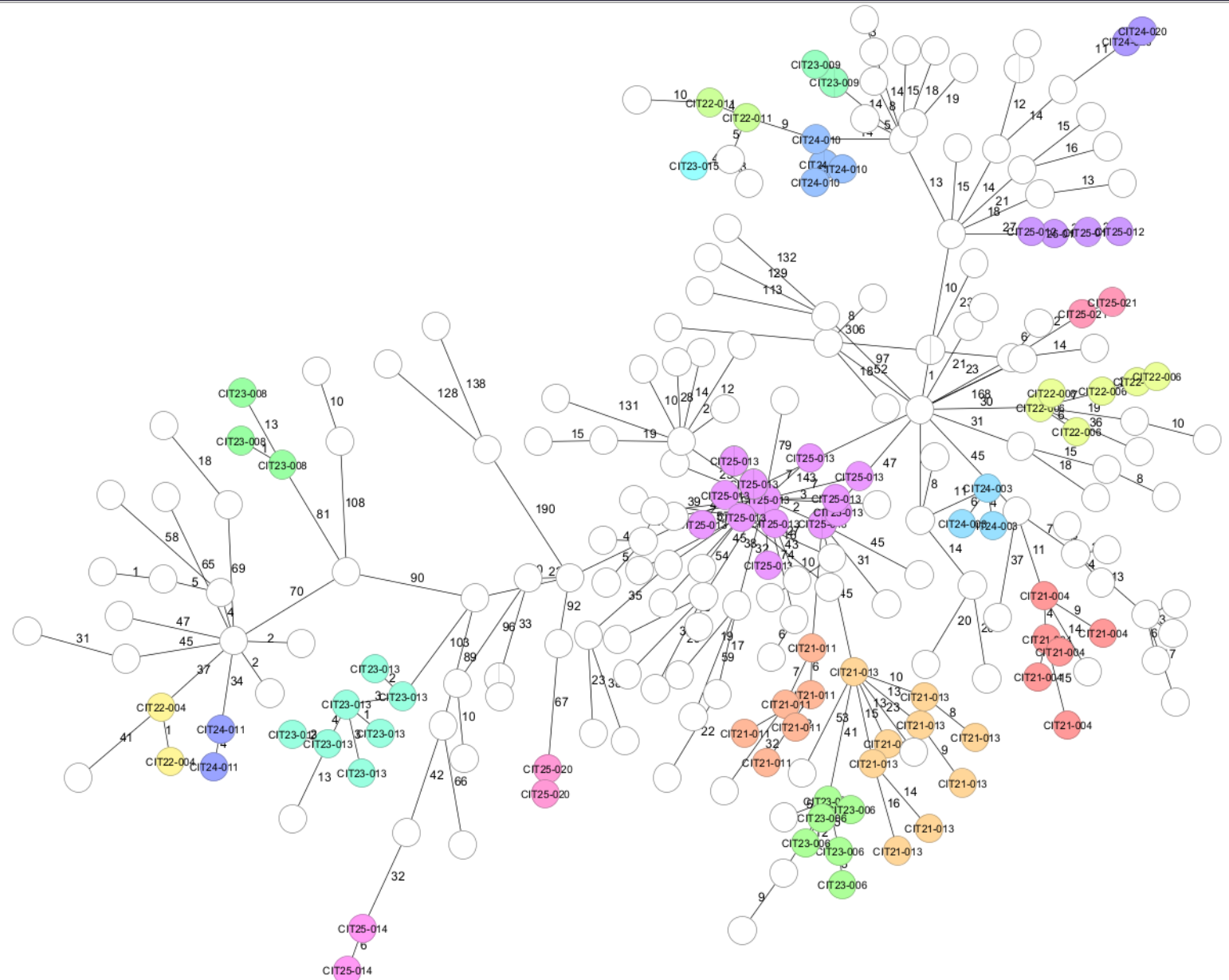
- EpiPulse Alert for a clone of *E.hormachei* ST1344 carrying carrying blaNDM-5
- We received 3 in a 6 week period, no further isolates have been seen
- 36 cases were identified in 6 EU/EEA countries
- No source identified as of yet- search continues

Irish CPE Clusters

ST22 *Citrobacter freundii* complex

Analysed using
cgMLST and
Accessory genomes.

Coloured by cluster

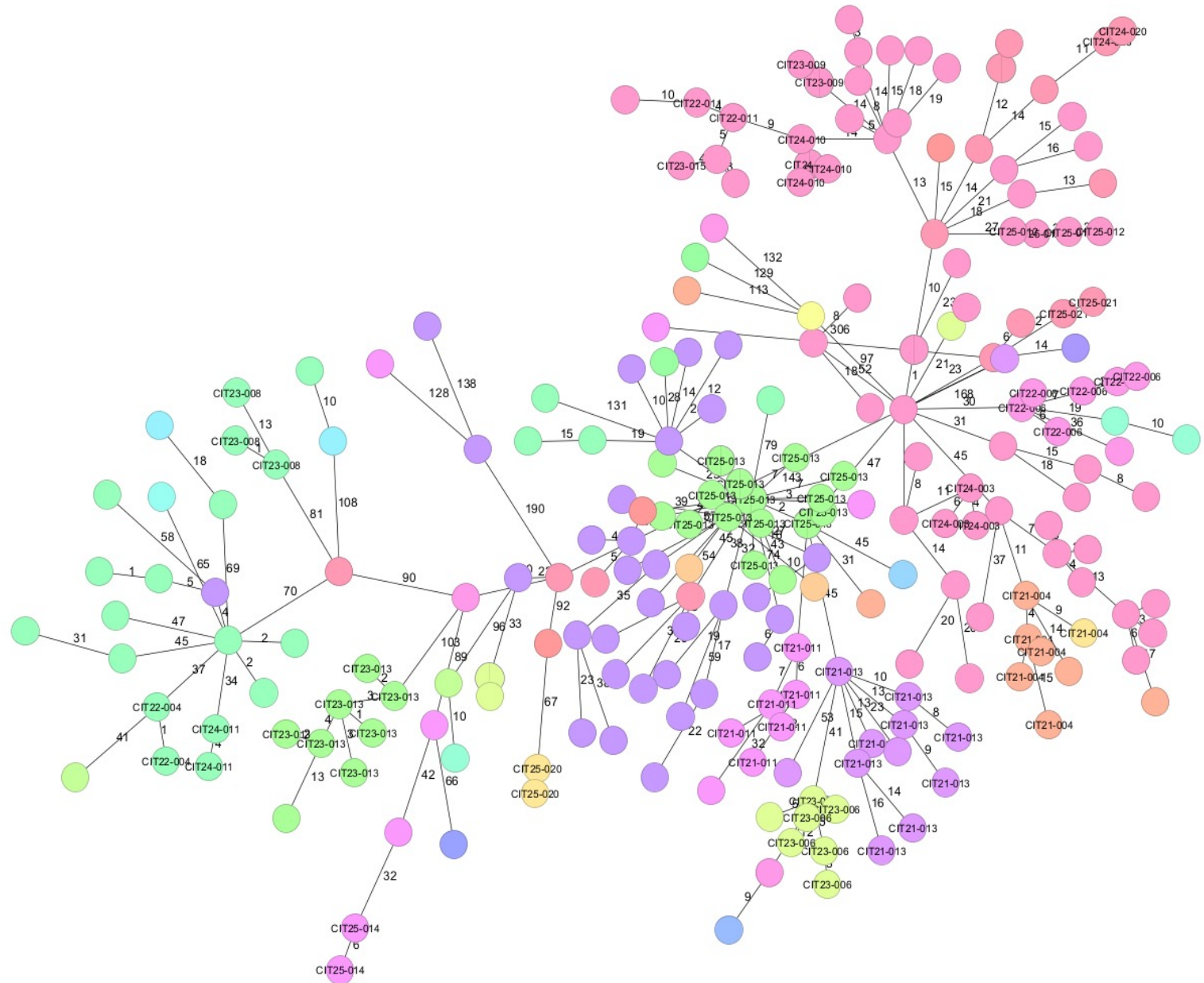


ST22 *Citrobacter freundii* complex

Analysed using
cgMLST and
Accessory genomes.

Coloured by hospital

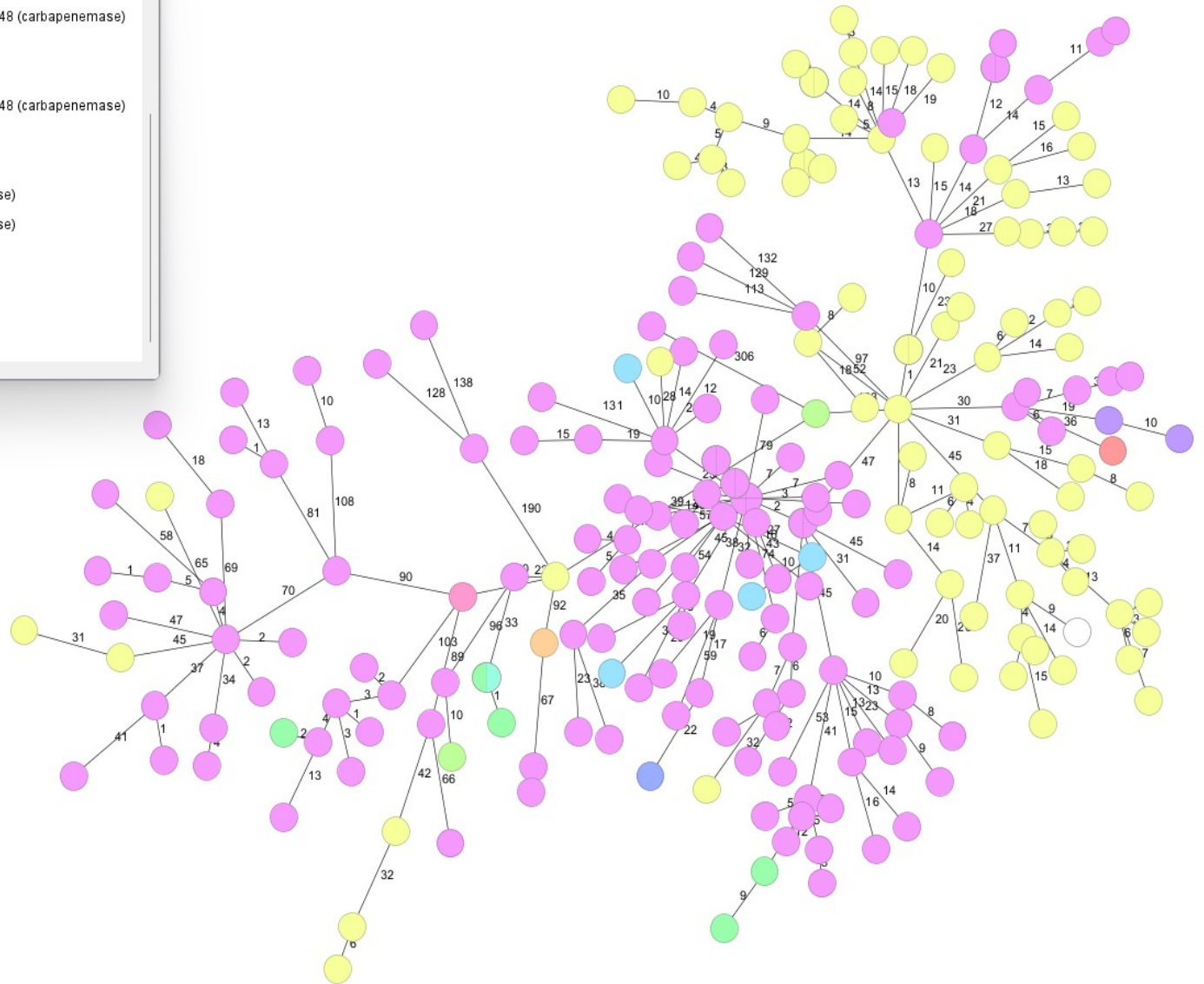
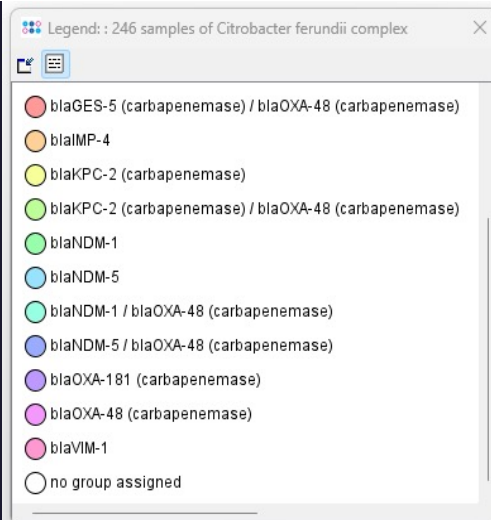
Most clusters in this example are limited to 1 hospital



ST22 *Citrobacter freundii* complex

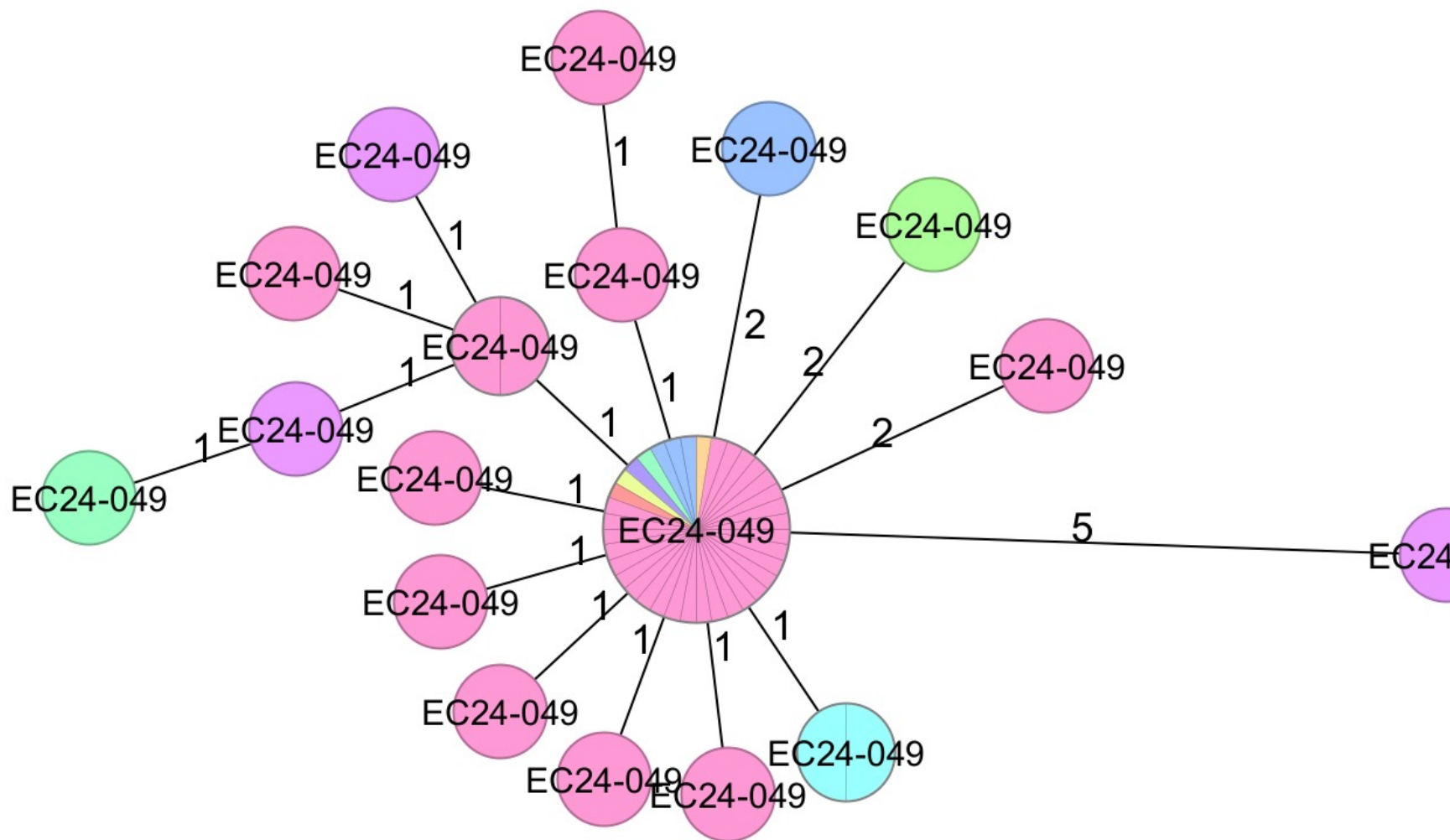
Analysed using
cgMLST and
Accessory
genomes.

Coloured by CPE
gene detected

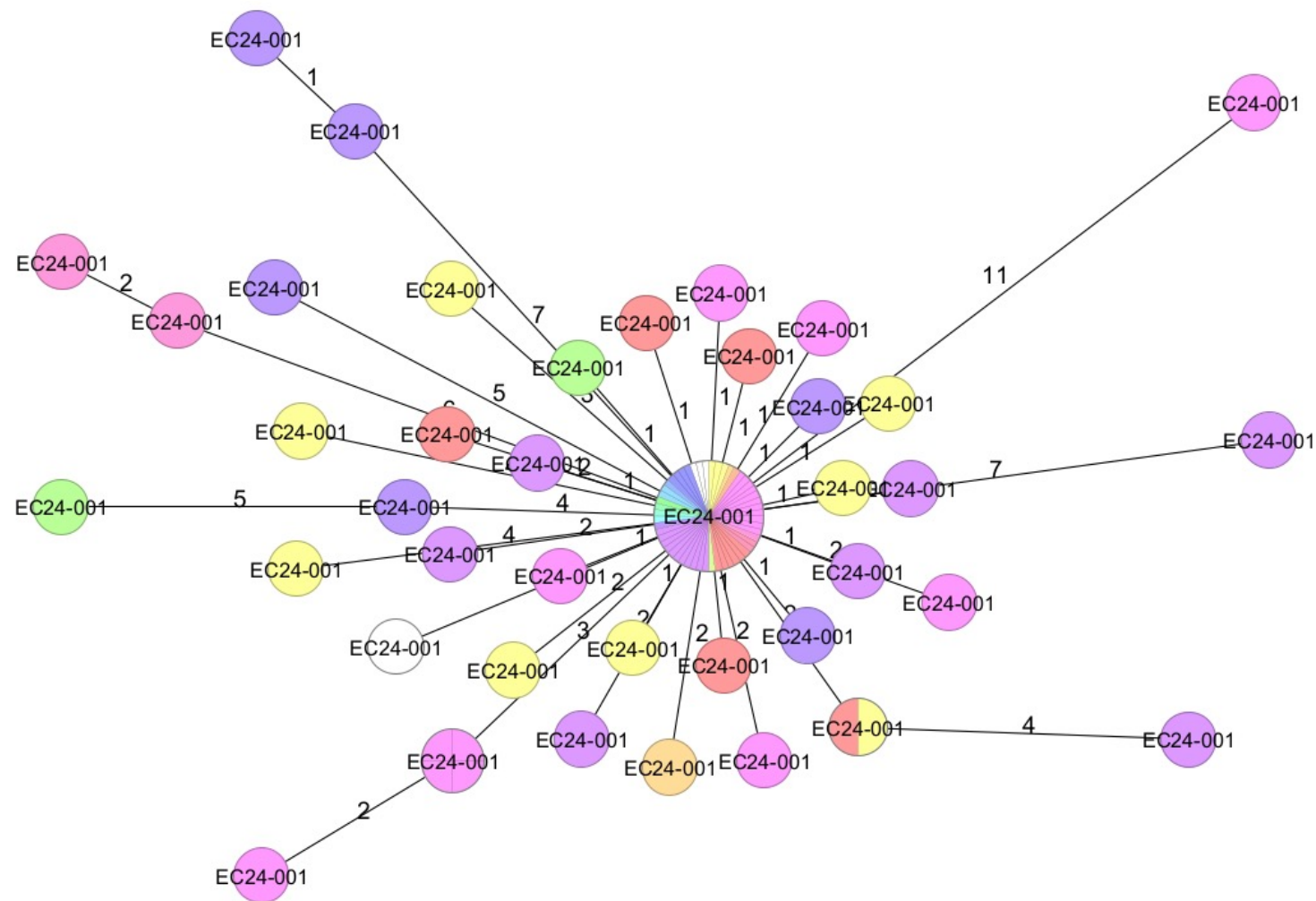


Inter Hospital clusters

This cluster contains 55 isolates from 10 different sites



24-001



NDM-1 Plasmid mediated outbreak

Long Reads used for plasmid Identification

ARGs present in plasmid:

Gene ID	E78938	E90850	E128606	E378246
<i>blaNDM-1</i>	✓	✓	✓	✓
<i>ble-MBL</i>	✓	✓	✓	✓
<i>mcr-9.1</i>	✓	✓	✓	✓
<i>catB3</i>	✓	✓	✓	✓
<i>blaOXA-1</i>	✓	✓	✓	✓
<i>aac(6')-Ib-D181Y</i>	✓	✓	✓	✓
<i>dfirA17</i>	✓	✓	✓	✓
<i>aadA5</i>	✓	✓	✓	✓
<i>sul1</i>	✓	✓	✓	✓
<i>mph(A)</i>	-	✓	✓	✓
<i>blaDHA-1</i>	-	✓	✓	✓
<i>qnrB4</i>	-	✓	✓	✓
<i>ere(A)</i>	-	✓	✓	✓
<i>arr-269927220</i>	-	✓	✓	✓
<i>aac(3)-II</i>	-	✓	✓	✓
<i>aac(6')-IIc</i>	-	✓	✓	✓
Total ARGs:	9	16	16	16

	E378246_p1	E90850_p1	E128606_p1	E78938_p1
E378246_p1	100.0	99.919312	99.936638	99.899689
E90850_p1	99.919312	100.0	99.975136	99.937073
E128606_p1	99.936638	99.975136	100.0	99.985748
E78938_p1	99.899689	99.937073	99.985748	100.0

All plasmids are conjugative of IncHI2A.

NDM plasmids of *E. cloacae* ST 88
E90850, *K. pneumoniae* ST 111
E128606, *E. cloacae* ST 1373 E378246
are identical. They all carry 16 ARGs

Data Source: University of Galway



Surveillance

National and International

National Surveillance

- Each *Salmonella*, *Shigella* and *Listeria* (SSL) isolate is reported to CIDR
- Each new SSL cluster is reported to the HPSC
 - Additions to clusters are also reported
 - HPSC Liaise with Public Health, questionnaires issued to patients.
- An EpiPulse event may be created if epidemiology is suggestive that this may be an international/travel related cluster this can be *Salmonella Shigella Listeria* or Carbapenemase Producing Organisms



Health Protection Surveillance Centre
Lárionad Faire um Chosaint Sláinte



National Surveillance

Concerning clusters may warrant further investigation or declaring an outbreak

Salmonella/Shigella/Listeria

- Liaise with DAFM colleagues if requested by HPSC to determine if they have related isolates.
- Outbreak management team meets and decides on further measures.
- Suspected outbreak isolates are prioritised for WGS and analysis.

CPE

- Liaise with hospitals affected
- Outbreak management team may meet in affected hospital and decide on further measures.
- Suspected outbreak isolates are prioritised for WGS and analysis.

National Surveillance

- Monthly line listings of CPE are sent to AMRIC, HPSC and to Hospitals
- Monthly line listings of SSL are sent to HPSC , Public Health and to Hospitals
- AMR Data sent quarterly for Salmonella and Shigella to HPSC
- They release data regularly via quarterly reports and annual reports.
- Both quarterly and annual reports share data on AMR amongst Salmonella and Shigella
- We also release annual reports on our activities

European Surveillance

- Comparing European outbreak sequences (EpiPulse events) to sequences in our own database.
- Responding to EpiPulse events.
- Informing the HPSC of any matching isolates.
- Real-Time reporting of our WGS data from Irish isolates to ECDC EpiPulse Cases system





EpiPulse

European surveillance
of CPE, *Salmonella*,
Shigella and *Listeria*

•Online portal for European public health authorities and other global partners

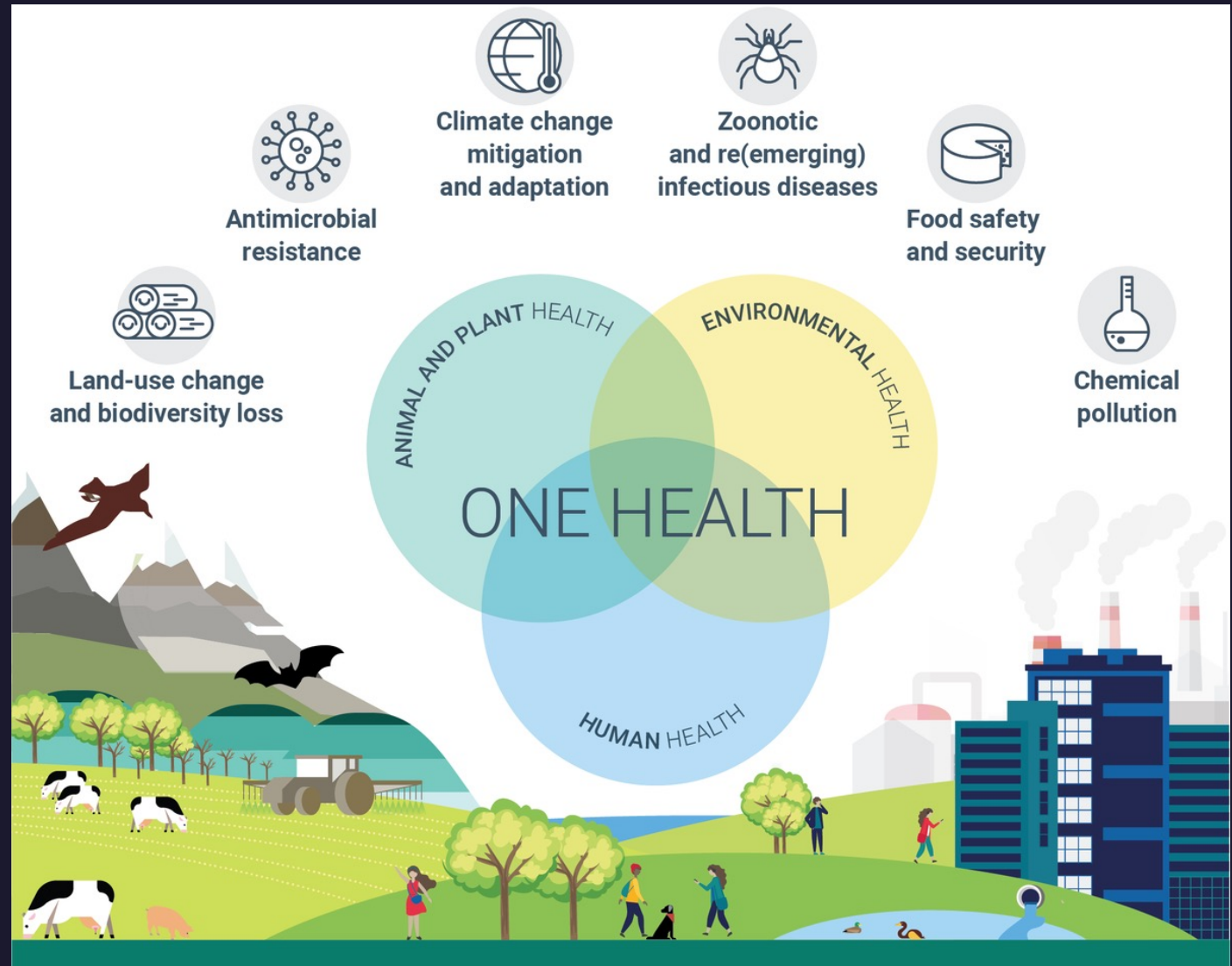
•Used to collect, analyse, share and discuss data for threat detection, monitoring, risk assessment and outbreak response.

•Indicator based and event based surveillance.

One Health

One Health is a multi-sectoral approach that aims to balance and optimise the **health of people, animals, plants, and their shared environment**, recognising their interconnection.

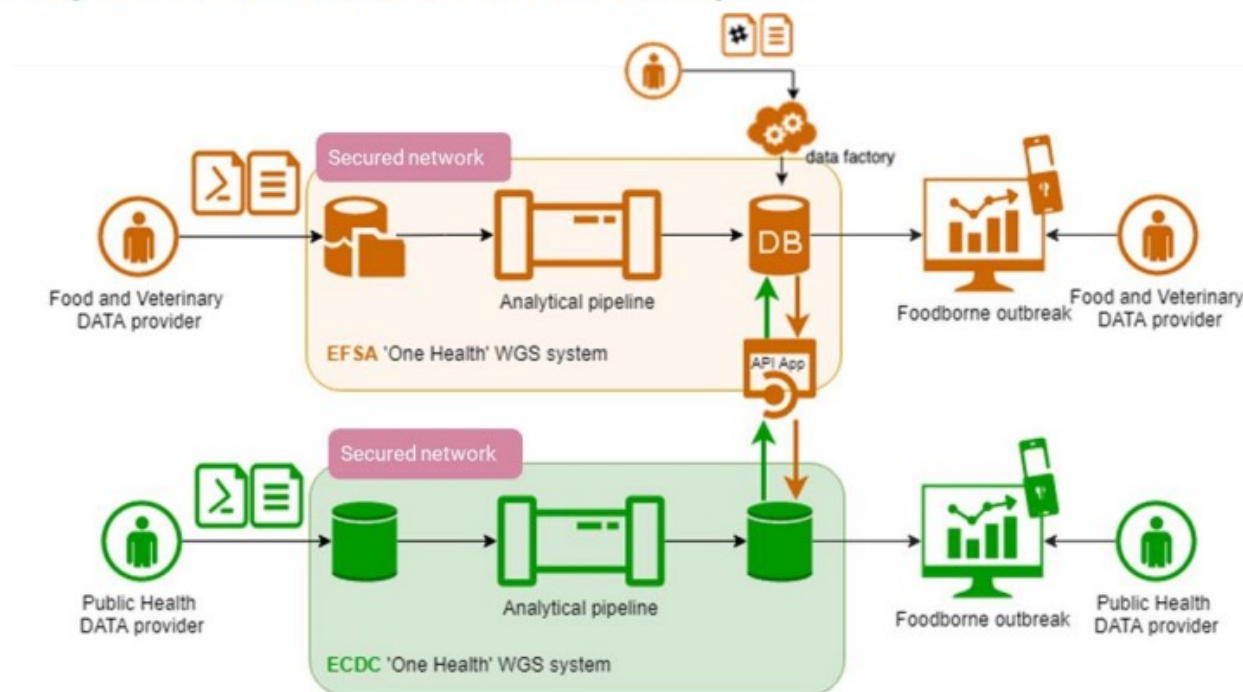
By promoting joint actions, One Health brings together professionals from various disciplines to tackle complex health threats in an integrated manner.



EU ONE HEALTH WGS DATA SHARING MODEL

COMBINATION OF APPROACHES

- **Controlled-Centralized Data Sharing at sector level:** EFSA and ECDC collect data centrally based on their remit
- **Cross-Sectoral Collaboration:** interoperability and comparability of the EFSA and ECDC platforms allow the data exchange for allowing cluster detection and outbreak investigation



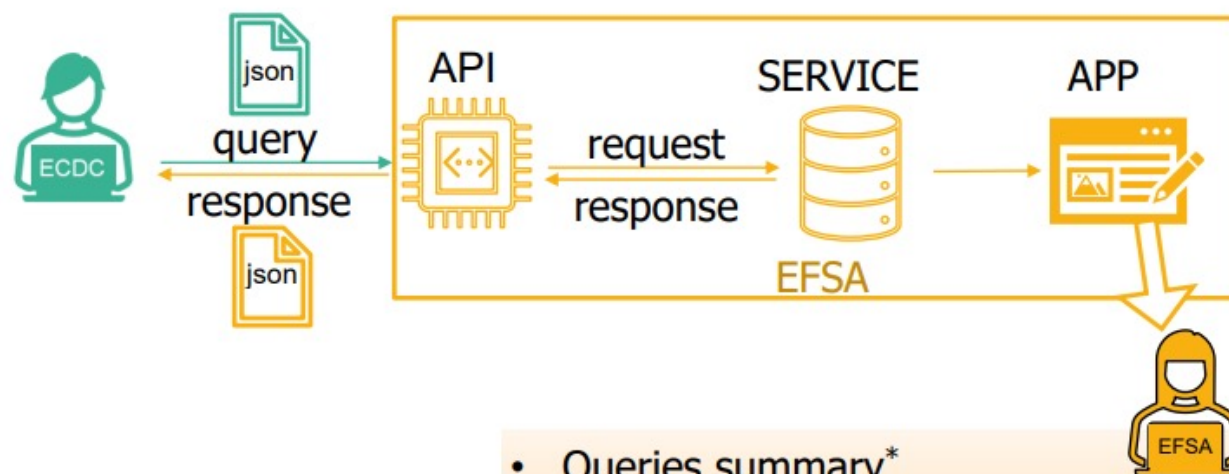
HOW ECDC/EFSA INTERACT: THE QUERY WORKFLOWS

Recurring weekly querying

- **Indicator-based surveillance**
- Automatic weekly querying
- Cluster definition at ECDC
- Fixed threshold

Ad hoc querying

- **Event-based surveillance**
- Querying on demand
- Linked to events in EpiPulse
- Multiple thresholds



- Queries summary*
- MSTs including human and food data

* >30k queries since 1st of July 2022
Average 2500 queries each month

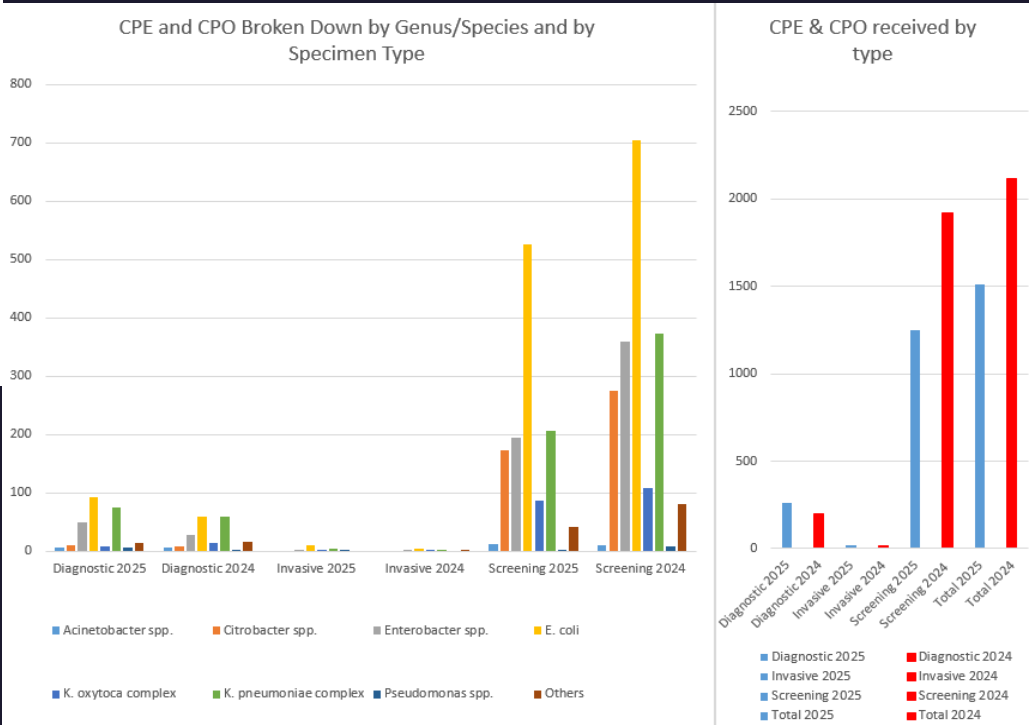


Annual Report Highlights

Species	Number of isolates					
	Diagnostic (including invasive) 2025	Diagnostic (including invasive) 2024	Diagnostic Invasive 2025	Diagnostic Invasive 2024	Rectal/ faeces Screening 2025	Rectal/ faeces Screening 2024
<i>Acinetobacter</i> spp.	6	6	0	0	13	11
<i>Citrobacter</i> spp.	10	9	0	0	174	275
<i>Enterobacter</i> spp.	49	29	3	3	195	360
<i>E. coli</i>	92	59	11	5	526	704
<i>K. oxytoca</i> complex	9	15	1	1	87	109
<i>K. pneumoniae</i> complex	75	60	4	3	206	373
<i>Pseudomonas</i> spp.	6	1	1	0	3	8
Others	14	17	0	2	42	81
Total CPE	249	189	19	14	1,229	1,902
Total CPO other than CPE¹	12	7	1	0	17	19
Total	261	196	20	14	1,246	1,921

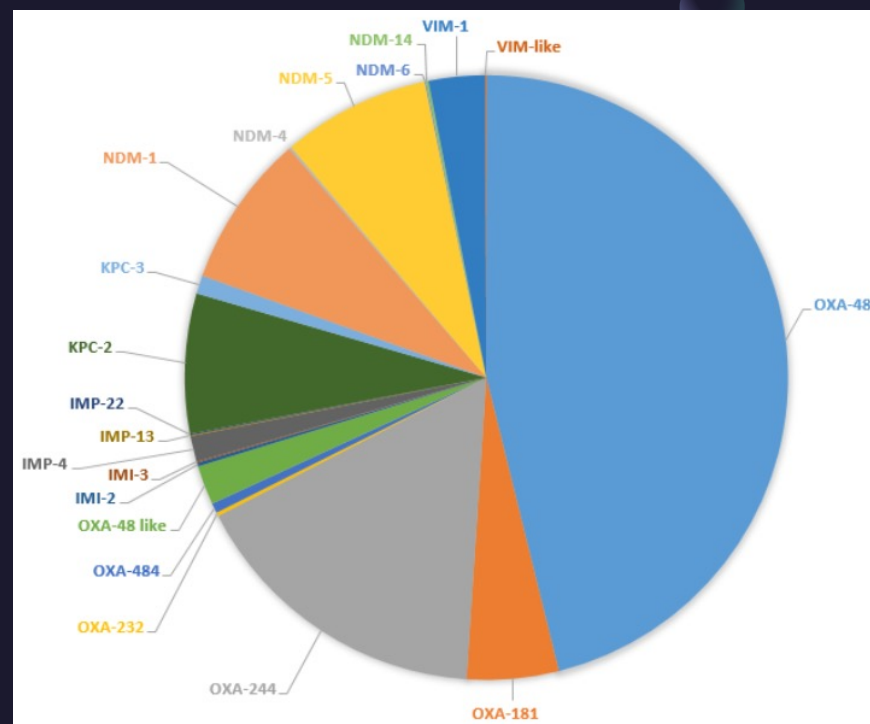
Typically diagnostic specimens make up 10% of our cases, as can be seen there was an increase of 33% on last year and a 43% increase in invasive diagnostic isolates

Screening samples were artificially lower due to the quota system imposed on all model 4 hospitals



CPE Trends- 2024 vs 2025

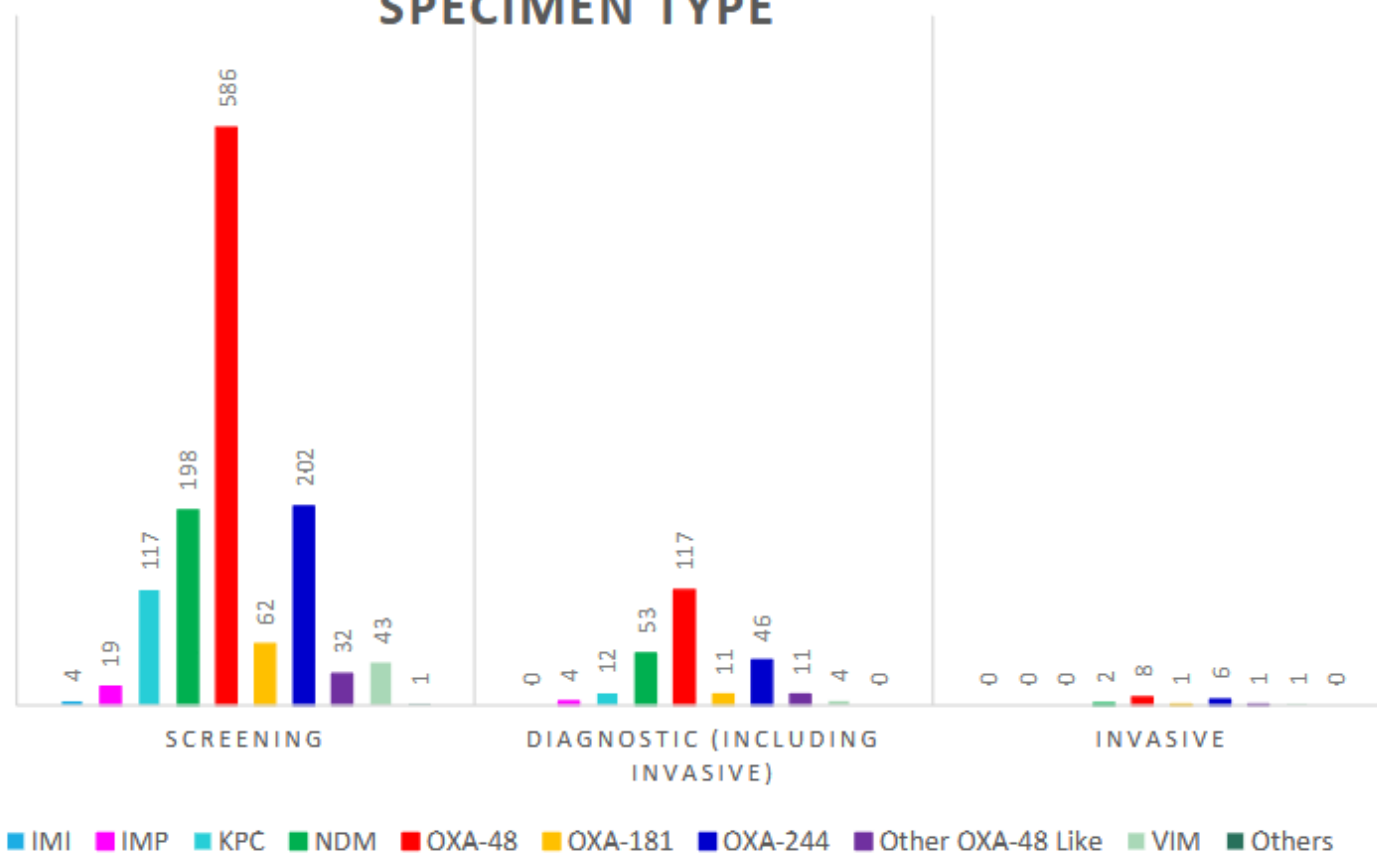
Carbapenemase Enzyme in Enterobacterales	Number of isolates					
	Diagnostic (including invasive) 2025	Diagnostic (including invasive) 2024	Diagnostic Invasive 2025	Diagnostic Invasive 2024	Rectal/faeces Screening 2025	Rectal/faeces Screening 2024
IMI	0	0	0	0	4	3
IMP	4	0	0	0	19	27
KPC	12	19	0	2	117	248
NDM	53	27	2	0	198	168
OXA-48	117	101	8	9	586	1077
OXA-181	11	12	1	1	62	83
OXA-244	46	27	6	2	202	238
Other OXA-48 Like	11	2	1	0	32	31
VIM	4	5	1	0	43	76
Others	0	0	0	0	1	1
Total	249	189	19	14	1,229	1,903
Isolates With >1 CPE enzyme found	9	4	0	0	35	49



NDM, IMP, OXA-48, OXA-244 continues to rise in diagnostic samples.

The only reductions we have seen are moderate reductions in KPC, OXA-181 and VIM from Diagnostic samples

CARBAPENEMASE ENZYME DETECTED BY SPECIMEN TYPE



Number of each type of carbapenemase enzyme detected broken down by variant identified for 2024 & 2025, from human isolates.

This includes Enterobacterales, Pseudomonas species and Acinetobacter species.

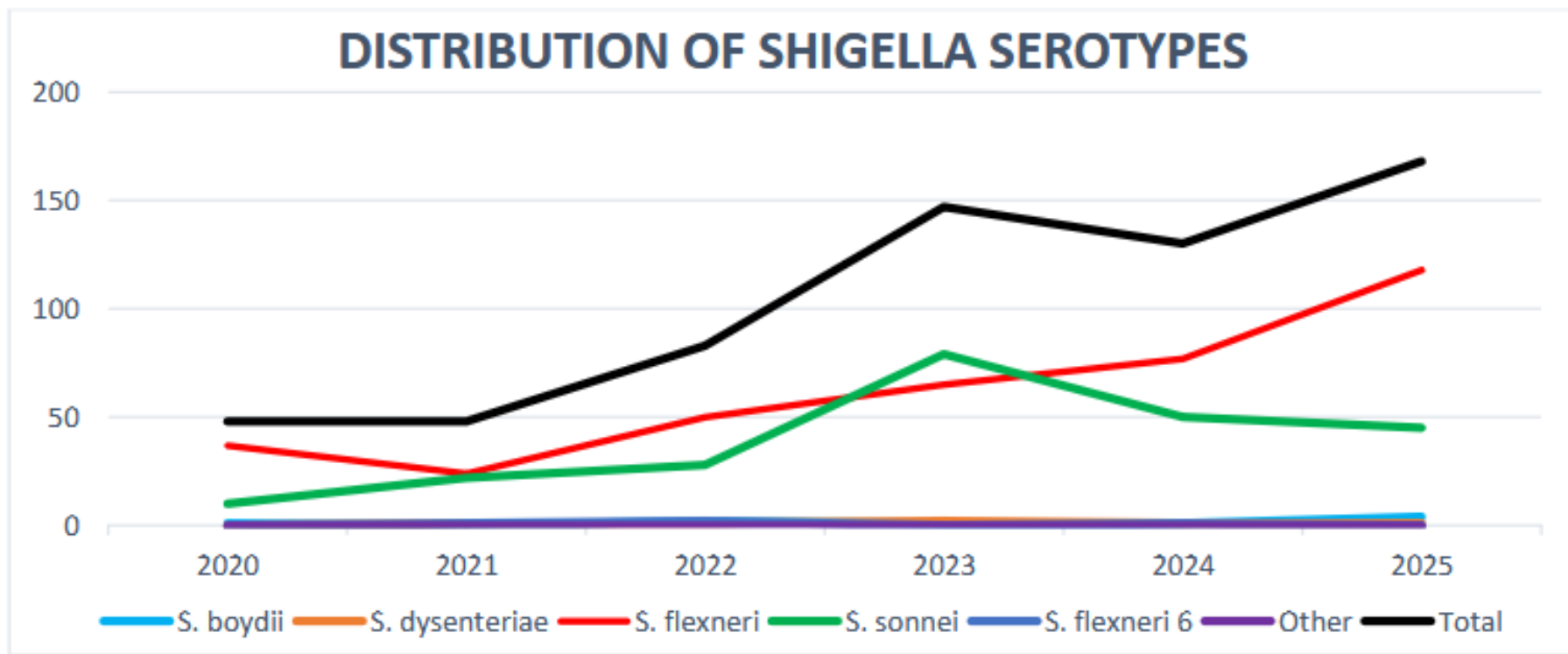
D= Diagnostic isolate
R= Rectal/screening isolate

Carbapenemase Gene Variants Detected in Enterobacterales in 2025														
OXA-48 Variants	OXA-48		OXA-181		OXA-244		OXA-48 like other							
	D	R	D	R	D	R	D	R						
2025	117	586	11	62	46	202	11	32						
2024	101	1077	12	83	27	238	2	31						
IMI Variants	IMI-2		IMI-3		IMI-4		IMI-6							
	D	R	D	R	D	R	D	R						
2025	0	3	0	1	0	0	0	0						
2024	0	0	0	0	0	1	0	2						
IMP Variants	IMP-1		IMP-4		IMP-13		IMP-22		IMP-like					
	D	R	D	R	D	R	D	R	D	R				
2025	1	0	4	17	0	1	0	1	0	1				
2024	0	2	0	25	0	1	0	1	0	0				
KPC Variants	KPC-2		KPC-3											
	D	R	D	R										
2025	9	105	3	12										
2024	10	187	9	61										

Number of each type of carbapenemase enzyme detected broken down by variant identified for 2024 & 2025, from human isolates continued.

NDM Variants	NDM-1		NDM-4		NDM-5		NDM-6		NDM-7		NDM-13		NDM-14	
	D	R	D	R	D	R	D	R	D	R	D	R	D	R
2025	27	109	1	1	28	94	1	0	0	0	0	0	0	2
2024	13	68	0	1	16	108	0	0	0	2	0	1	0	0
VIM Variants	VIM-1		VIM-2		VIM-5		VIM-like							
	D	R	D	R	D	R	D	R						
2025	5	43	0	1	0	0	0	1						
2024	4	76	0	3	1	0	0	0						
Others	OXA-23		OXA-58		GES-11		GES-24		FRI-8					
	D	R	D	R	D	R	D	R	D	R				
2025	6	6	0	2	0	1	0	1	0	0				
2024	5	2	0	1	0	0	0	0	0	1				

Shigella Trends 2020-2026

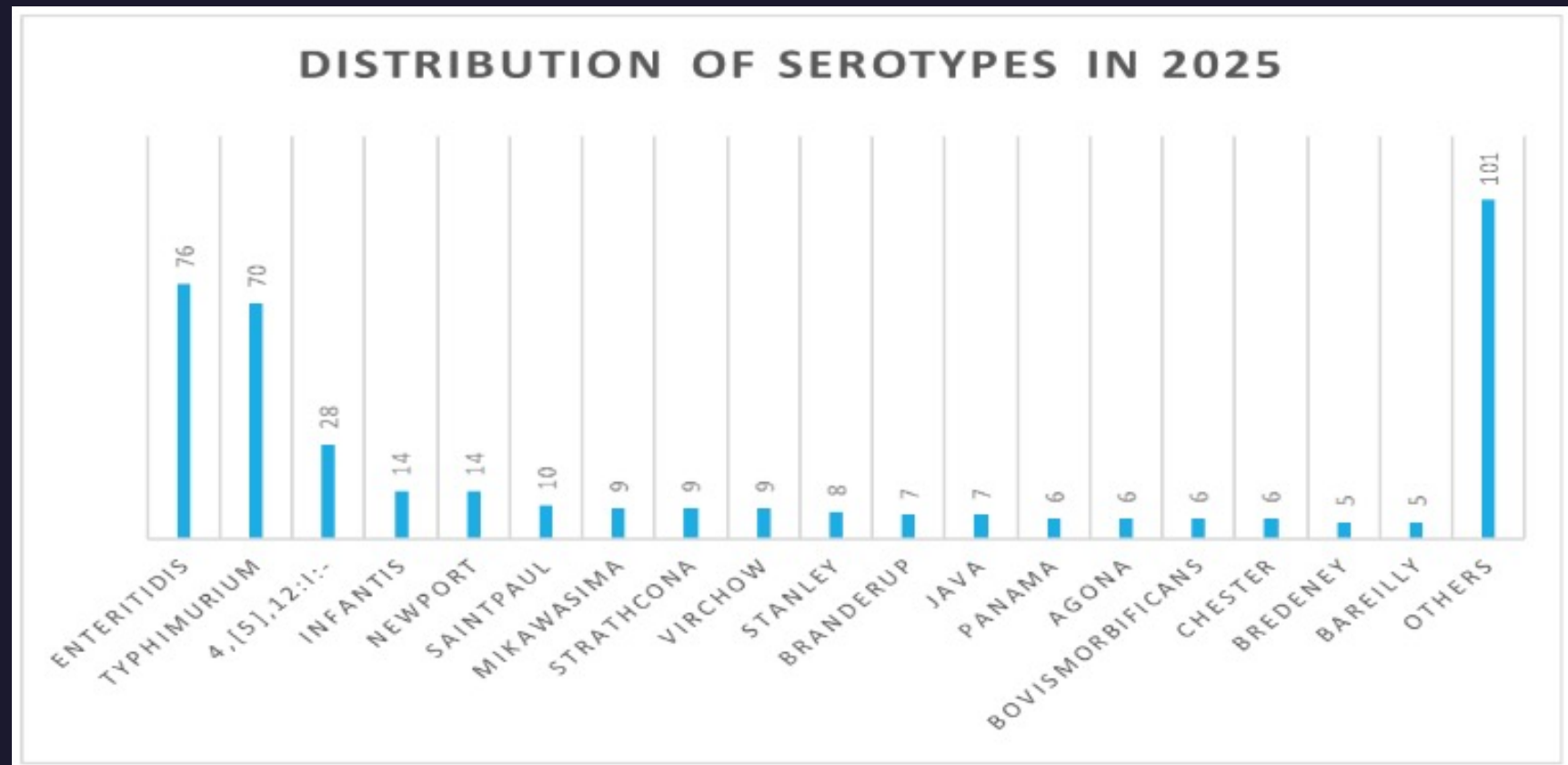


AMR Trends in Shigella

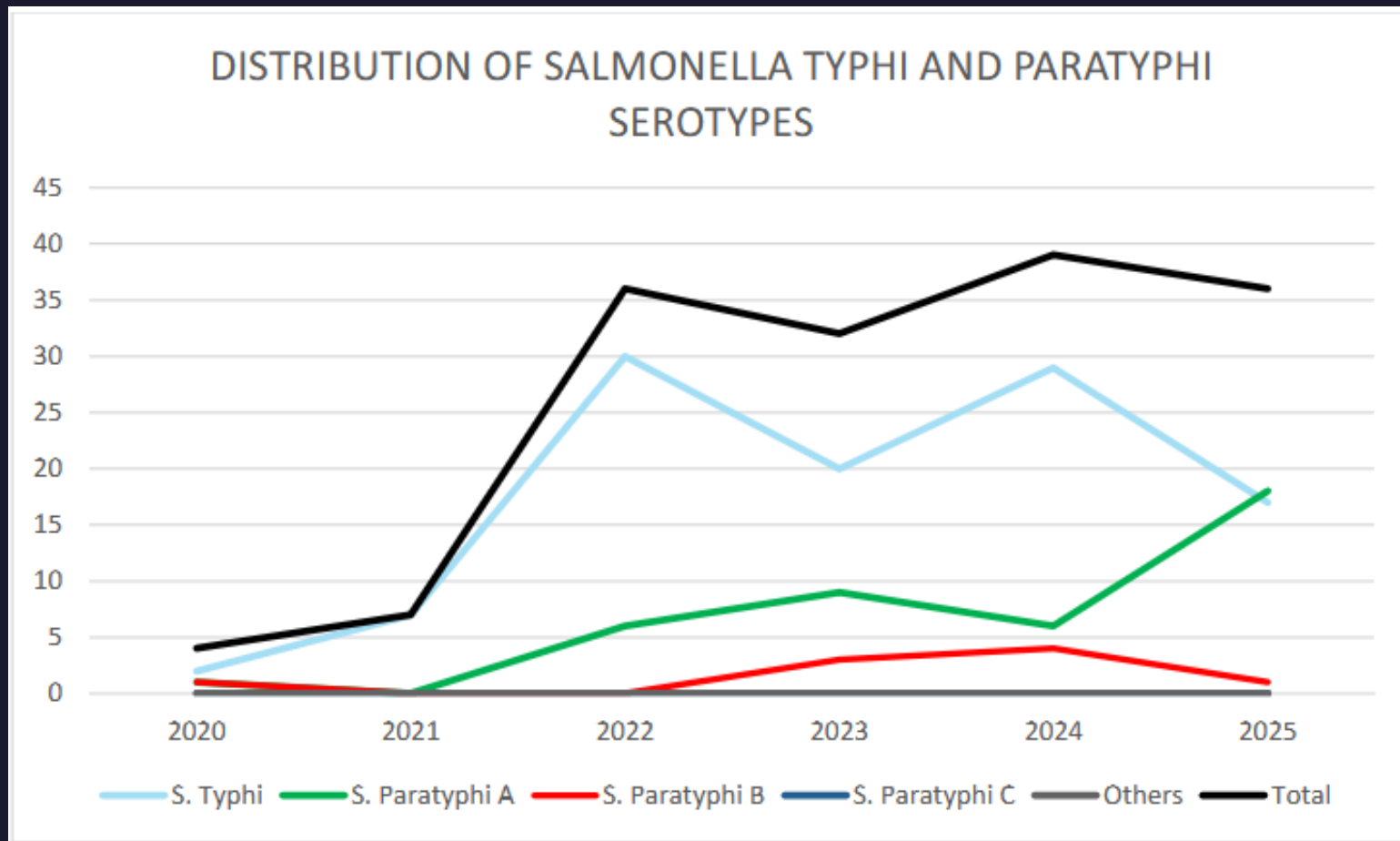
Serotype	% Resistance (or number when total isolates is less than 10)				
	Amoxicillin	Azithromycin	Ceftriaxone	Ciprofloxacin	Meropenem*
<i>S.boydii</i>	1/4	0/4	0/4	2/4	0/4
	25%	0%	0%	50%	0%
<i>S.dysenteriae</i>	1/1	0/1	0/1	1/1	0/1
	100%	0%	0%	100%	0%
<i>S.flexneri</i>	116/118	72/118	17/118	69/118	1/118
	98%	61%	14.4%	58%	<1%
<i>S.sonnei</i>	35/45	22/45	30/45	35/45	0/45
	78%	49%	67%	78%	0%
All Shigella	155/168	94/168	140/168	107/168	1/168
	92%	56%	83%	64%	<1%

* 1 *S.flexneri* was received to the laboratory that harboured blaOXA-181 which was susceptible to meropenem on phenotypic testing.

Non-Typhoidal Salmonella

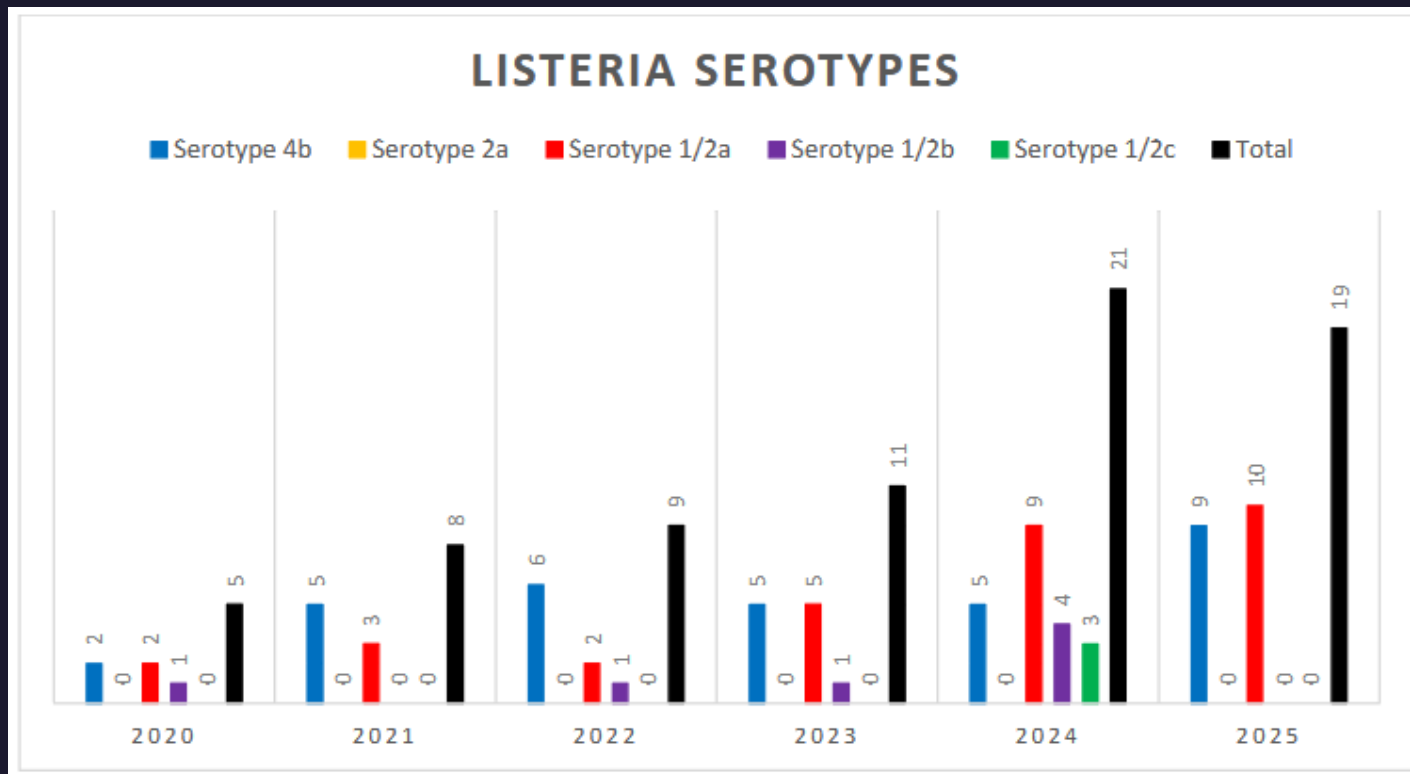


Typhoidal Salmonella Trends 2020-2026

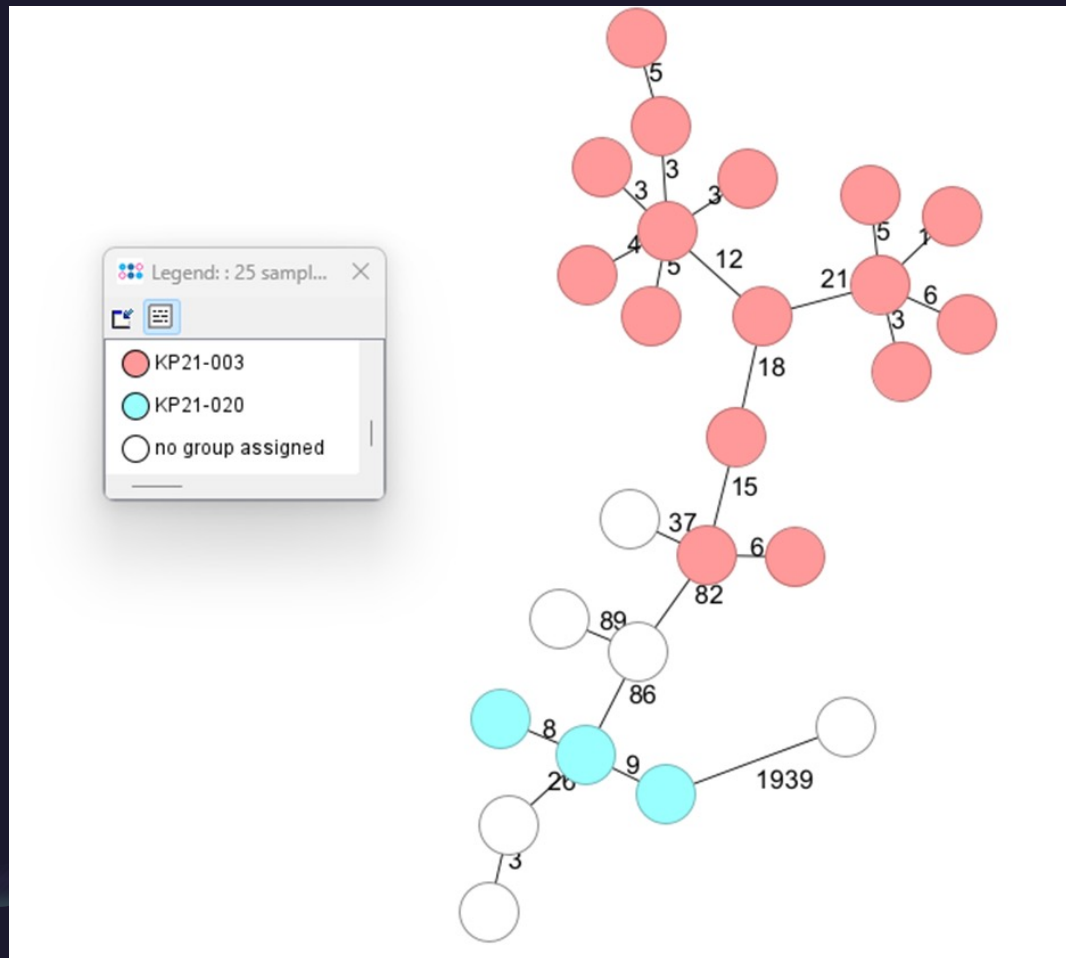


Listeria Tends 2020-2025

The increases in 2024 and 2025 was largely due to cluster L24-002



Hypervirulent *K.pneumonia* ST 23 in Ireland



Annual ST23 hvKP Ireland

2019 - 20

2020 - 20

2021 - 10

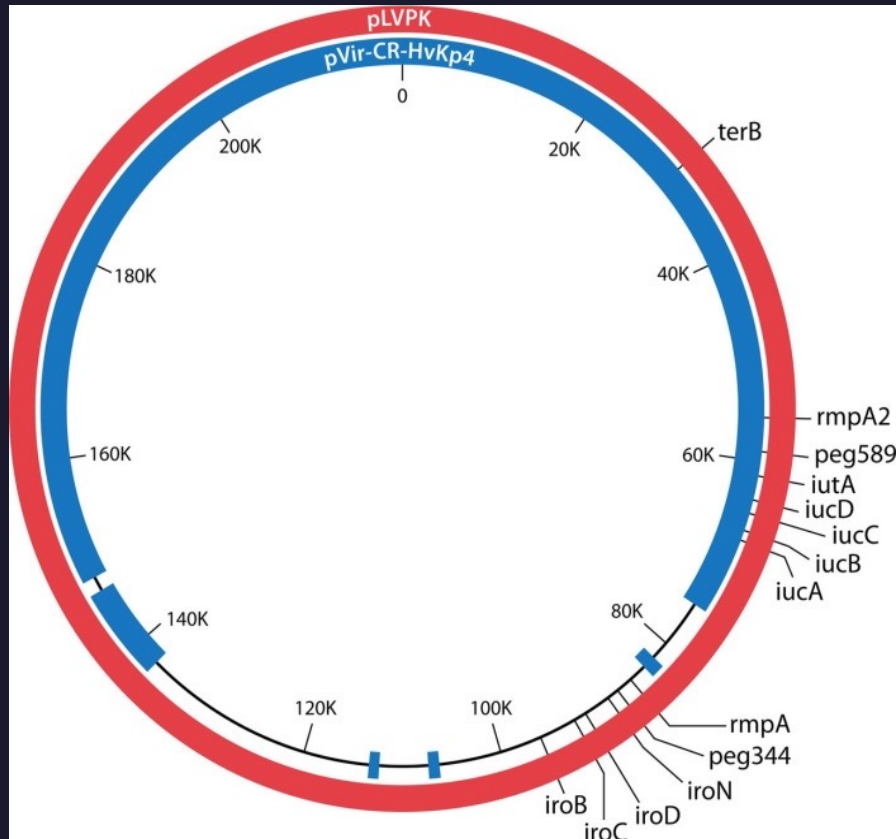
2022 - 32

2023 - 15

2024 - 17

It is not known how many ST23 hvKp that are not CPE are being missed

Virulence detection



Hypervirulent *K.pneumoniae* Kleborate scoring guide:

- 1 → Yersiniabactin (*ybt*) only
- 2 → Yersiniabactin (*ybt*) / Colibactin (*clb*) without aerobactin
- 3 → Aerobactin (*iucA*) only
- 4 → Aerobactin (*iucA*) and yersiniabactin (*ybt*)
- 5 → Aerobactin, (*iucA*) yersiniabactin (*ybt*) and colibactin (*clb*)

This isolate has a Kleborate/ECDC score of 0-2 therefore is NOT a hypervirulent strain of *Klebsiella pneumoniae*.

This isolate has a Kleborate/ECDC score of 3-5 therefore is a hypervirulent strain of *Klebsiella pneumoniae*.

Other hvKP Markers

Other virulence genes are often seen in the pVIR plasmid and can be used to determine hypervirulence

Some groups use these 5 biomarkers to define hvKP:

- *iucA* - Aerobactin
- *iroB* - Salmochelin
- *rmpA* - Regulator of mucoid phenotype
- *rmpA2* - Activator for capsular polysaccharide (CPS) synthesis
- *peg-344* - Metabolite transporter

Neither scheme is perfect on its own, most likely will be a mix of the two that yields the most accurate results

CPE Reports

CLINICIAN / DESTINATION Test Doctor Testing HCO Laboratory UCHG	SURNAME APEX10	FORENAMES CHANGEOFNAME	BOARD No. / I.D. No. APEX10	
	ADDRESS Changeofaddress changeofaddress	changeofaddress	SEX M	DATE OF BIRTH 10/06/2014
	SPECIMEN Isolate urine	LAB No.		
WARD / CLINIC	DATE & TIME OF COLLECTION 11/12/2018 09:00	DATE & TIME RECEIVED 15/12/2018 07:00	ME230001F	

Identification

Isolate identified as **Escherichia coli**

Whole Genome Sequencing

Sequence type

Sequence Type 10

Relevant Resistance genes

Carbapenemase Producing Enterobacterales (CPE) Confirmed

Relevant Resistance Genes

Carbapenemase genes

ESBL genes

ampC genes

+/- mcr gene

Porin mutations causing Carbapenem resistance

blaOXA-48 gene (CPE) Detected
blaCTX-M-15 gene (ESBL) Detected
blaOXA-10gene(non-CPE related) Detected
blaTEM-1C gene (B-lactamase) Detected
blaSHV-12 gene (B-lactamase) Detected

Relevant Plasmids

- OXA-48 IncL or IncM(pOXA-48)
- OXA-181 IncX3

Relevant Plasmids

IncL/M (pOXA-48) plasmid Detected

AST result

Colistin M.I.C. 0.5 ug/ml
Interpretation : Susceptible
Ceftazadime-avibactam MIC <=1 ug/ml
Interpretation : Susceptible

The above MICs were determined by broth microdilution.

AST for Clinical isolates

Colistin only on MBL's

CPE Reports

Identification

Isolate identified as **Klebsiella pneumoniae**

Whole Genome Sequencing

Sequence type

Sequence Type 23

This isolate is part loose KP21-003

Relevant Resistance genes

Carbapenemase Producing Enterobacterales (CPE) Confirmed

blaOXA-48 gene (CPE) Detected

blaSHV-11 gene (B-lactamase) Detected

mcr genes NOT detected

Relevant Plasmids

IncL/M (pOXA-48) plasmid Detected

Plasmid group A3

Kleborate
vhKP scoring

This isolate has a Kleborate/ECDC score of 3-5 therefore is a hypervirulent strain of *Klebsiella pneumoniae*.

This report on sequence type (ST), resistance genes and plasmids is generated using Bioinformatics software SeqSphere (version 10.5.4) supplemented by external software as required.
This is a non-accredited test.

Report from Carbapenemase Producing Enterobacterales Reference Laboratory.

Please note that the Board / I.D.No. is the External Reference No
Please record date and time of collection on request form.

End of Report

Shigella Reports

Sequence Type

245

This isolate is part of loose cluster SH19-012

Sequence Type

+/- Cluster code

Predicted Serotype

Shigella flexneri

Predicted Serotype

Relevant Resist. Determinants

blaOXA-1 gene (Amoxicillin) Detected

mph(A) gene (Azithromycin) Detected

catA1 gene (Chloramphenicol) Detected

tet(B) gene (Tetracycline) Detected

Relevant Resistance Genes

Sequence Type (ST), predicted serotype and relevant resistance genes are determined by analysis of Whole Genome Sequencing (WGS).

WGS Analysis is performed using SeqSphere Software version 10.5.4

WGS data is continuously analysed for similarities between isolates and if any evidence of outbreaks or linked cases are detected then the relevant public health authorities will be informed.

This is a non-accredited test.

End of Report

Salmonella Reports

Sequence Type 11
This isolate is part of cluster S25-015.

Sequence Type
+/- Cluster code

Predicted Serotype
Salmonella Enteritidis

Predicted Serotype

Relevant Resist. Determinants
gyrA mutation(s) (Quinolone) Detected

Relevant Resistance Genes

Sequence Type (ST), predicted serotype and relevant resistance genes are determined by analysis of Whole Genome Sequencing (WGS).

WGS Analysis is performed using SeqSphere Software version 10.5.4

WGS data is continuously analysed for similarities between isolates and if any evidence of outbreaks or linked cases are detected then the relevant public health authorities will be informed.

This is a non-accredited test.

End of Report

Listeria Reports

Sequence Type 37
This isolate is part of cluster L24-002

Sequence Type
+/- Cluster code

Predicted Serotype
L.monocytogenes 1/2a

Predicted Serotype

Relevant Resist. Determinants
None Detected

Relevant Resistance Genes

Sequence Type (ST), predicted serotype and relevant resistance genes are determined by analysis of Whole Genome Sequencing (WGS).

WGS Analysis is performed using SeqSphere Software version 10.5.4

WGS data is continuously analysed for similarities between isolates and if any evidence of outbreaks or linked cases are detected then the relevant public health authorities will be informed.

This is a non-accredited test.

End of Report

Considerations for laboratories

- Quality Assurance

- How will you monitor quality?
 - External quality assessment, internal quality controls, inter laboratory comparison are some options
- Will you be going for accreditation?

- Sequencing platform(s)

In house sequencing or send out?

If in house:

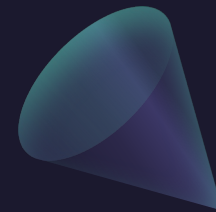
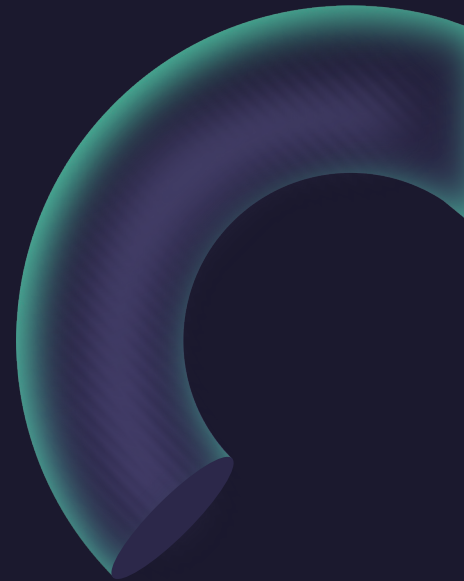
- Cost, environmental impact, facilities available.
- Ease of use, integration into current IT systems
- What do you require? Long reads, short reads or hybrid assembly?
- Cost benefits- can you avoid doing two types of sequencing if long reads are
- IVDR regulations



Considerations for laboratories

- **Bioinformatic analysis:**

- Commercial analysis package vs in house pipelines?
- What will you do with the data? Individual reports, quarterly reports, annual reports?
 - What do you want to achieve with the data?
- What will you put in your reports? Do you give all the info you have or do you use a curated approach for simplified easy to read reports?
- Data storage, cloud based or local storage.



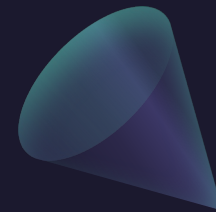
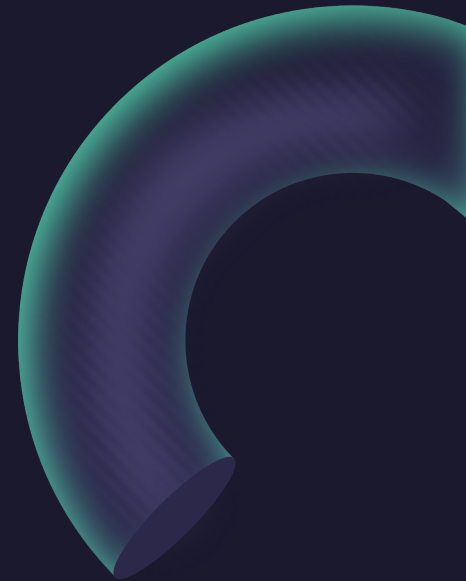
Future Developments for GRLS

- Long Read Sequencing

- We are in the process of procuring a long read sequencer

- Plasmid analysis and clustering

- We currently only have half the story with CPE, while we can easily detect and monitor clonal outbreaks we cannot comment on plasmid mediated outbreaks at present.



Thank you for listening!

Galway Reference Laboratory
Service Team:

- Christina Clarke
- Alma Tuohy
- Ger Doran
- Paula Ward
- Feena Cunnane
- Anne Coleman
- Prof. Martin Cormican



References/Further info can be found at

HSE-Health Protection Surveillance Centre. Provisional annual report on Gastroenteric and Zoonotic Diseases in Ireland, 2024. Dublin, HPSC, April 2025.

[https://www.hpsc.ie/a-z/gastroenteric/gastroenteritisoriid/surveillancereports/2024%20Gastroenteric%20and%20Zoonotic%20diseases%20Slideset_\(Provisional%20trends%20to%20the%20end%20of%20Q4%202024\)_v1.0.pdf](https://www.hpsc.ie/a-z/gastroenteric/gastroenteritisoriid/surveillancereports/2024%20Gastroenteric%20and%20Zoonotic%20diseases%20Slideset_(Provisional%20trends%20to%20the%20end%20of%20Q4%202024)_v1.0.pdf)

HSE-Health Protection Surveillance Centre. Quarterly report on Gastroenteric and Zoonotic Diseases in Ireland, Q1 2025. Dublin, HPSC, May 2025

<https://www.hpsc.ie/a-z/gastroenteric/gastroenteritisoriid/surveillancereports/Q1%202025%20Gastroenteric%20and%20Zoonotic%20diseases%20Slideset.pdf>

GALWAY REFERENCE LABORATORY REPORT 2024-

<https://www.saolta.ie/sites/default/files/publications/Galway%20Reference%20Laboratory%20Annual%20Report%202024%20Final.pdf>

GALWAY REFERENCE LABORATORY REPORT 2019-2023-

https://www.saolta.ie/sites/default/files/publications/Galway%20Reference%20Laboratory%20Annual%20Report%202019-2023_Final.pdf

EFSA AND ECDC ONE HEALTH WGS SYSTEM power point [EFSAhttps://www.efsa.europa.eu/sites/default/files/2023-09/presentation-1-rossi-nannapaneni.pdf](https://www.efsa.europa.eu/sites/default/files/2023-09/presentation-1-rossi-nannapaneni.pdf)

Illumina Website has more info, for Sequencing workflow video see <https://www.youtube.com/watch?v=EDVKxSNdSic>

ECDC: <https://www.ecdc.europa.eu/en/publications-data/epipulse-european-surveillance-portal-infectious-diseases>

AMRIC Monthly report August 2025 [https://www.hpsc.ie/a-](https://www.hpsc.ie/a-z/microbiologyantimicrobialresistance/carbapenemresistantenterobacteriaceae/surveillancereports/Mch%2008%20AMRIC%20August%202025%20CPE%20monthly%20report.pdf)

[z/microbiologyantimicrobialresistance/carbapenemresistantenterobacteriaceae/surveillancereports/Mch%2008%20AMRIC%20August%202025%20CPE%20monthly%20report.pdf](https://www.hpsc.ie/a-z/microbiologyantimicrobialresistance/carbapenemresistantenterobacteriaceae/surveillancereports/Mch%2008%20AMRIC%20August%202025%20CPE%20monthly%20report.pdf)

Insights into recurring multi-country outbreaks of *Salmonella* Strathcona associated with tomatoes, Europe, 2011 to 2024 <https://www.eurosurveillance.org/content/10.2807/1560-7917.ES.2025.30.41.2500224>